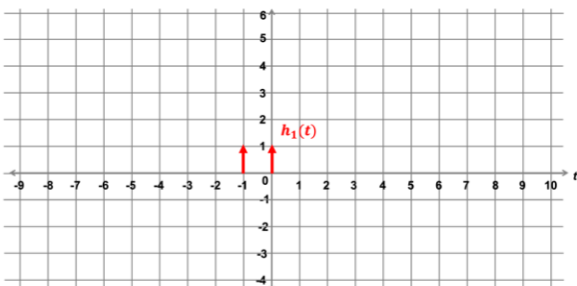
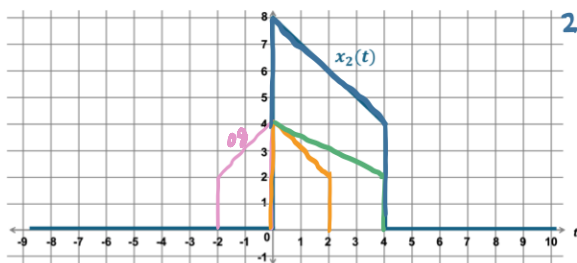
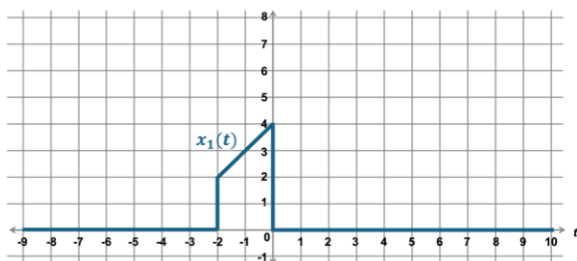


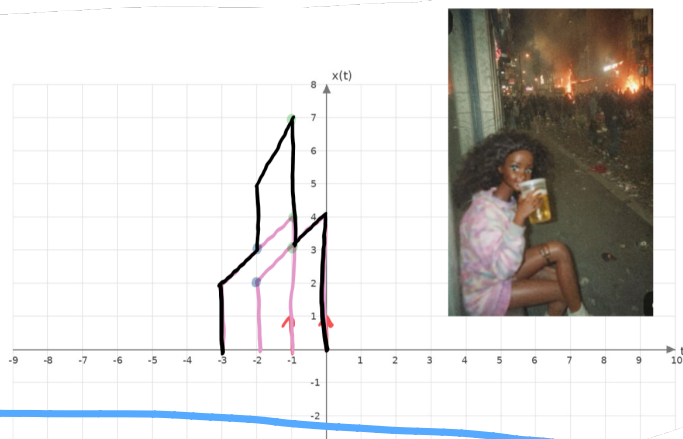
1. (15 points) Waveforms and Convolution

The signals $x_1(t)$, $x_2(t)$, and $h_1(t)$ are as graphed below.



- a. (7 points) Write the signal $x_2(t)$ in terms of $x_1(t)$. Hint: $x_2(t)$ is a transformed version of $x_1(t)$
 b. (8 points) Graph the output of the convolution: $y_1(t) = x_1(t) * h_1(t)$.

Note that the heights of the delta function arrows indicate their amplitudes.



3. (10 points) Laplace Transform & Its Inverse

- a. (5 points) Determine the Laplace transform $X(s)$ of the signal:

$$x(t) = e^{-j7t}u(t-1)$$

$$\begin{aligned} x(t) &= e^{-j7(t-1+1)}u(t-1) \\ &= e^{-j7} [e^{-j7(t-1)}u(t-1)] \\ &= e^{-j7} \left[\frac{e^{-s}}{s+j7} \right] \end{aligned}$$



- b. (5 points) Determine the inverse Laplace Transform of $H(s) = \frac{s^2+10s+18}{s^2+7s+12}$.

$$\begin{aligned} H(s) &= \frac{s^2+10s+18}{s^2+7s+12} \\ &= \frac{s^2+7s+12}{s^2+7s+12} + \frac{3s+6}{s^2+7s+12} \\ &= 1 + 3 \left[\frac{A}{(s+3)} + \frac{B}{(s+4)} \right] \\ &= 1 + 3 \left[\frac{A}{(s+3)} + \frac{B}{(s+4)} \right] \\ &= 1 + \frac{6}{s+4} - \frac{3}{s+3} \\ &= \delta(t) + 6e^{-4t}u(t) - 3e^{-3t}u(t) \end{aligned}$$

Q5 (possible)

$$a_n = \frac{42}{8} \int_0^4 (2-t) \cos(n\omega_0 t) dt$$
 b/c even can now multiply by 2
 y-intercept
 slope of line

- d. (5 points) Write the formula for $V_{out}(s)$ in terms of $V_{in}(s)$. Your answer should be written in terms of $A_1, A_2, A_3, A_4, I_1, I_2, V_{in}(s), V_{out}(s)$.

2 x (-t)

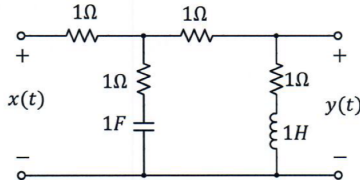
$$\Delta = A_1 A_4 - A_3 A_2 \Rightarrow I_2 = \frac{\Delta I_2}{\Delta}$$

$$\Delta I_2 = A_1 V_2 - A_3 V_1 = \frac{A_1 V_2 - A_3 V_1}{A_1 A_4 - A_3 A_2}$$

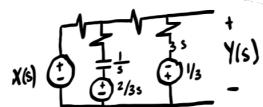
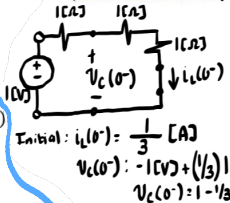
$$V_{out}(s) = (s+2) I_2 = \frac{1}{4}$$

4. (20 points) Applications of the Laplace Transform

Consider the following circuit the input $x(t) = 1V$ for $t < 0$, and $x(t) = (1 - e^{-t+1})u(t-1)$ for $t > 0$.



- a. (2 points) Sketch the Laplace-domain version of the given circuit with the initial conditions indicated (use the back sheet for your work as needed and write the final answer below).



- b. (3 points) Write the Laplace transform of the input for $t \geq 0$ and write the final answer below).

$$\begin{aligned} x(t) &= (1 - e^{-t+1})u(t-1) \\ &= u(t-1) - e^{-t+1}u(t-1) \\ &= u(t-1) - e^{-(t-1)}u(t-1) \\ &\Rightarrow \frac{e^{-s}}{s} - \frac{e^{-s}}{s+1} \\ X(s) &= e^{-s} \left(\frac{1}{s} - \frac{1}{s+1} \right) \end{aligned}$$



3. (10 points) Laplace Transform & Its Inverse

- a. (5 points) Determine the Laplace transform $X(s)$ of the signal:

$$x(t) = e^{-j7t}u(t-1)$$

$$\begin{aligned} x(t) &= e^{-j7(t-1+1)}u(t-1) \\ &= e^{-j7} [e^{-j7(t-1)}u(t-1)] \\ &= e^{-j7} \left[\frac{e^{-s}}{s+j7} \right] \end{aligned}$$

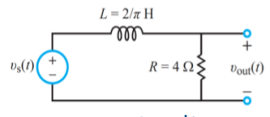


- b. (5 points) Determine the inverse Laplace Transform of $H(s) = \frac{s^2+10s+18}{s^2+7s+12}$.

$$\begin{aligned} H(s) &= \frac{s^2+10s+18}{s^2+7s+12} \\ &= \frac{s^2+7s+12}{s^2+7s+12} + \frac{3s+6}{s^2+7s+12} \\ &= 1 + 3 \left[\frac{A}{(s+3)} + \frac{B}{(s+4)} \right] \\ &= 1 + \frac{6}{s+4} - \frac{3}{s+3} \\ &= \delta(t) + 6e^{-4t}u(t) - 3e^{-3t}u(t) \end{aligned}$$

6. (20 points) Applications of Fourier Methods

a. (5 points) Calculate the output $v_{out}(t)$ generated by the following RL circuit when the input signal is a triangular waveform. The Fourier series for the triangular waveform is given below.

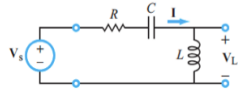

$$x(t) = \sum_{n=-\infty}^{\infty} \frac{8A}{n^2 \pi^2} \cos\left(\frac{2n\pi t}{T_0}\right)$$
$$V_{out} = \frac{R}{R+j\omega L} V_s$$
$$\omega_0 = \frac{2\pi}{T_0} \Rightarrow H(\omega) = \frac{4}{4+jn\omega_0 \frac{2}{\pi}}$$
$$= \frac{4}{4+jn \frac{2\pi}{T_0} \cdot \frac{2}{\pi}} = \frac{1}{1+j\frac{n}{T_0}} \quad (A)$$

$$|H(n\omega_0)| = \frac{1}{\sqrt{1+(\frac{n}{T_0})^2}}$$
$$\angle H(n\omega_0) = 0 - \tan^{-1}\left(\frac{n}{T_0}\right)$$
$$\Rightarrow y(t) = \sum_{n=-\infty}^{\infty} \frac{8A}{n^2 \pi^2} \cdot \frac{1}{\sqrt{1+(\frac{n}{T_0})^2}} \cdot \cos\left(\frac{2n\pi t}{T_0} - \tan^{-1}\left(\frac{n}{T_0}\right)\right)$$



Grading:
• 1 pt trying (A)
• +2 pt correct $H(\omega)$ (A)
• +1 pt (B)
• +1 pt (C)

b. (5 points) What type of filter does the following circuit implement? If $L = 0.1H$ and $C = 10\mu F$, calculate the center frequency ω_0 and the value of R that will achieve a quality factor $Q = 10$.


$$H(\omega) = \frac{j\omega L}{R + \frac{1}{j\omega C} + j\omega L} = \frac{-\omega^2 LC}{j\omega RC + (1 - \omega^2 LC)}$$
$$|H(\omega)| = \frac{\omega^2 LC}{\sqrt{(1 - \omega^2 LC)^2 + (\omega RC)^2}}$$

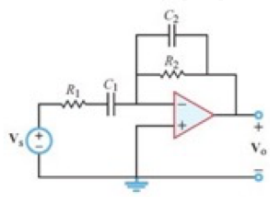
as $\omega \rightarrow 0$ $|H(\omega)| \rightarrow 0$
as $\omega \rightarrow \infty$ $|H(\omega)| \rightarrow 1$
when $1 - \omega^2 LC = 0$ or $\omega = \frac{1}{\sqrt{LC}}$, denominator is minimum $\Rightarrow$ peak response

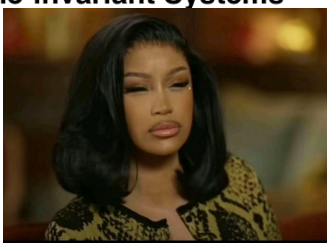
Grading:
• 1 pt for trying (A, B, C, D)
• +1 pt each (A, B, C, D)

(B) This is a high-pass filter with $\omega_0 = \frac{1}{\sqrt{LC}}$
 $\omega_0 = \frac{1}{\sqrt{0.1 \times 10 \times 10^{-6}}} = 10^3 \text{ rad/sec}$ (C)
 $Q = \frac{1}{R} \sqrt{\frac{L}{C}} = 10 \Rightarrow \frac{1}{R} \sqrt{\frac{0.1}{10 \times 10^{-6}}} = 10 \Rightarrow R = 10 \Omega$ (D)

2. (10 points) Linear Time-invariant Systems

Consider the following op-amp circuit:





a. (4 points) Derive an expression for the magnitude of the transfer function $|H(\omega)|$

$$H(\omega) = - \frac{R_2 \parallel \frac{1}{j\omega C_2}}{R_1 + \frac{1}{j\omega C_1}} = \frac{-j\omega R_2 C_1}{(1 + j\omega R_1 C_1)(1 + j\omega R_2 C_2)}$$
$$|H(\omega)| = \frac{\omega R_2 C_1}{\sqrt{(1 + (\omega R_1 C_1)^2) \sqrt{(1 + (\omega R_2 C_2)^2)}}}$$

1st order numerator, 2nd order denominator

b. (3 points) What type of filter does this circuit represent? Explain your answer.

It is non zero between these extremes

Grading:
• 1
• 2

$\Rightarrow$ Band-pass filter.

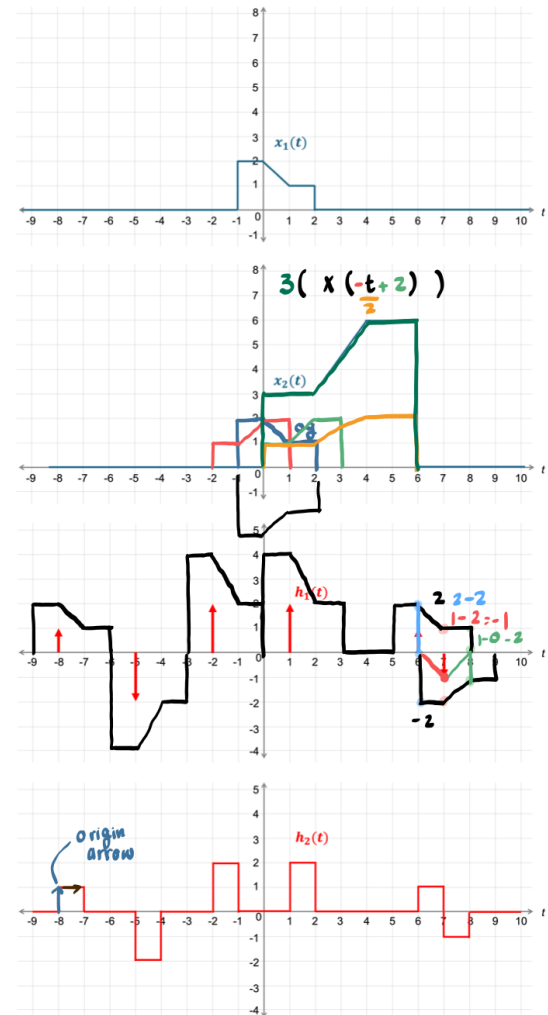
c. (3 points) Identify the poles and zeros for this system.

zero @ $\omega_1 = \frac{1}{R_2 C_1}$ (A)

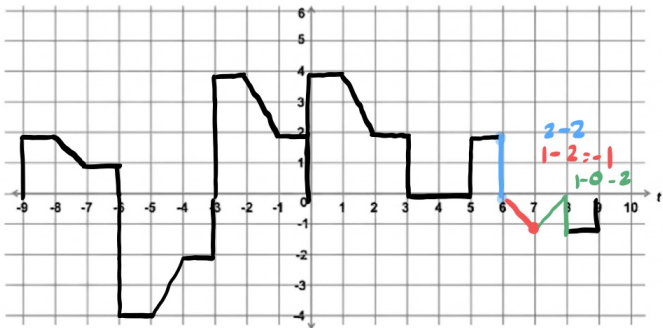
poles @ $\omega_2 = \frac{1}{R_1 C_1}$ (B) & $\frac{1}{R_2 C_2} = \omega_3$ (C)

1. (15 points) Waveforms and Convolution

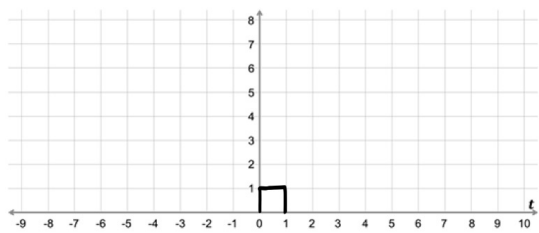
The signals $x_1(t)$, $x_2(t)$, $h_1(t)$, and $h_2(t)$ are as graphed below.



b. (6 points) Graph the output of the convolution: $y_1(t) = x_1(t) * h_1(t)$, where $*$ represents convolution. Note that the heights of the delta function arrows indicate their amplitudes. Your final answer must be presented in the space below, and this will be the only answer graded.

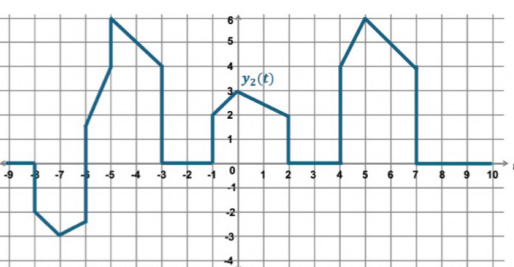
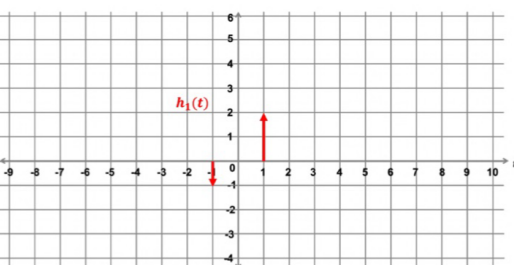
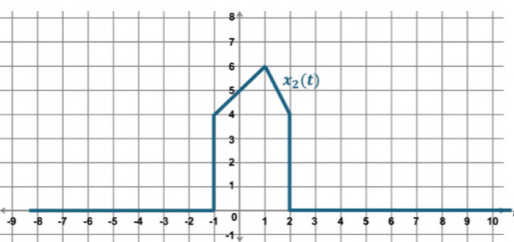
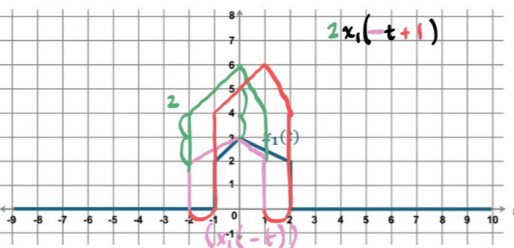


c. (5 points) If the signal $h_2(t)$ graphed above satisfies the equation: $h_2(t) = z(t) * h_1(t)$, where $*$ represents convolution, then graph the signal $z(t)$. Your final answer must be presented in the space below, and this will be the only answer graded.

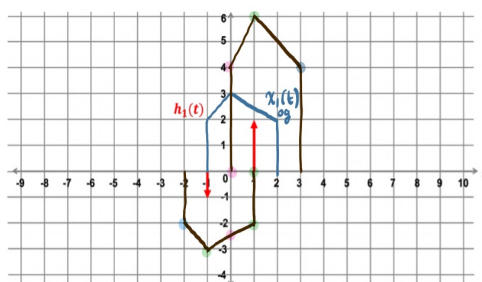


1. (15 points) Waveforms and Convolution

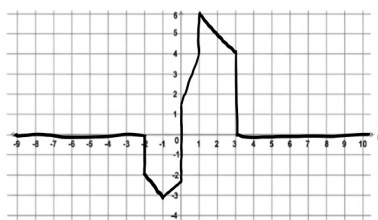
The signals $x_1(t)$, $x_2(t)$, $h_1(t)$, and $y_2(t)$ are as graphed below.



(6 points) Graph the output of the convolution: $y_1(t) = x_1(t) * h_1(t)$, where $*$ represents convolution. Note that the heights of the delta function arrows indicate their amplitudes. Your final answer must be presented in the space below, and this will be the only answer graded.



(5 points) If the signal $y_2(t)$ graphed above satisfies the equation: $y_2(t) = x_1(t) * z(t)$, where $*$ represents convolution, then graph the signal $z(t)$. Your final answer must be presented in the space below, and this will be the only answer graded.



3. (10 points) Laplace Transform & Its Inverse

a. (5 points) Determine the Laplace transform $X_3(s)$ of the signal:

$$x_3(t) = (t-1)^2 e^{-(t-1)} u(t-1)$$

Consider the signal: $e^{-t}u(t)$

$$e^{-t}u(t) \leftrightarrow \frac{1}{s+1} \quad \text{--- (A)}$$

$$te^{-t}u(t) \leftrightarrow -\frac{d}{ds}\left(\frac{1}{s+1}\right) = \frac{1}{(s+1)^2} \quad \text{--- (B)}$$

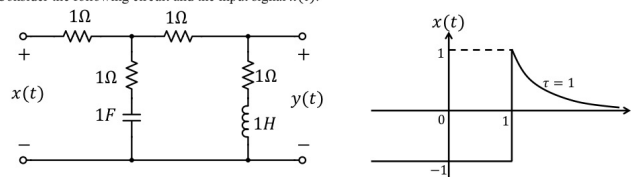
$$t^2 e^{-t}u(t) \leftrightarrow -\frac{d}{ds}\left(\frac{1}{(s+1)^2}\right) = \frac{2}{(s+1)^3} \quad \text{--- (C)}$$

Using the time-shifting property

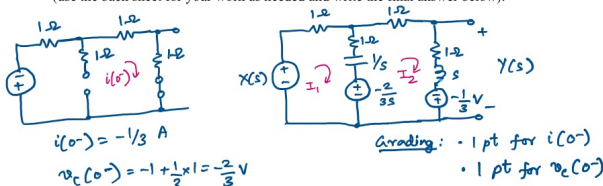
$$x_3(t) = (t-1)^2 e^{-(t-1)} u(t-1) \leftrightarrow \frac{2}{(s+1)^3} e^{-s} \quad \text{--- (D)}$$

4. (20 points) Applications of the Laplace Transform

Consider the following circuit and the input signal $x(t)$.



a. (2 points) Sketch the Laplace-domain version of the given circuit with the initial conditions indicated (use the back sheet for your work as needed and write the final answer below).



b. (3 points) Write the Laplace transform of the input for $t \geq 0$ (use the back sheet for your work and write the final answer below).

$$\text{For } t \geq 0, x(t) = -u(t) + u(t-1) + e^{-(t-1)}u(t-1)$$

$$\Rightarrow X(s) = -\frac{1}{s} + \frac{e^{-s}}{s} + e^{-s} \cdot \frac{1}{s+1}$$

Grading: 1 pt attempt (min)
1 pt each for (A), (B), (C)

c. (5 points) Calculate the Laplace-domain output $Y(s)$ for $t \geq 0$. (use the back sheet for your work and write the final answer below). Your answer should be in the form of a ratio of two polynomials.

KVL:

$$I_1 \left(1 + \frac{1}{s}\right) - I_2 \left(1 + \frac{1}{s}\right) = X(s) + \frac{2}{3s} \quad \text{--- (A)}$$

$$-I_1 \left(1 + \frac{1}{s}\right) + I_2 \left(\frac{1}{s} + 1 + 1 + s\right) = \frac{2}{3s} - \frac{1}{3} \quad \text{--- (B)}$$

$$\Rightarrow \begin{vmatrix} \frac{2s+1}{s} & -\frac{(s+1)}{s} \\ -\frac{(s+1)}{s} & \frac{(s^2+3s+1)}{s} \end{vmatrix} = \frac{2s^2+6s+3}{s}$$

$$\Delta_{I_2} = \begin{vmatrix} \frac{2s+1}{s} & \frac{3s^2+2}{3s} \\ -\frac{(s+1)}{s} & -\frac{(s+2)}{3s} \end{vmatrix} = -\frac{(2s+3)}{3s} + \frac{(s+1)X(s)}{s}$$

$$Y(s) = I_2(1+s) - \frac{1}{3} = \frac{\Delta_{I_2}(s+1) - \frac{1}{3}}{\Delta}$$

$$= \frac{1}{3} \cdot \frac{s}{2s^2+6s+3} (s+1) - \frac{1}{3}$$

$$= \frac{s(s+1) - (2s^2+6s+3)}{3(2s^2+6s+3)} = \frac{-(s^2+5s+3)}{3(2s^2+6s+3)}$$

d. An LTI system has the following differential equation description, with initial conditions $x(0^-) = 1$, $y(0^-) = 1$, $x'(0^-) = 1$, $y'(0^-) = 1$, and input $x(t) = e^{-3t}u(t)$:

$$\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 10y = 2\frac{dx}{dt} + x$$

(i) (3 points) Identify the modes of this system.

$$\text{characteristic equation: } s^2 + 7s + 10 = 0 \Rightarrow (s+5)(s+2) = 0$$

$$\text{poles: } \{-5, -2\} \quad \text{modes: } \frac{1}{s+5}, \frac{1}{s+2}$$

Grading: 1 pt for each mode

(ii) (8 points) Calculate the natural response of this system in the Laplace domain.

Convert the LCDE to Laplace domain:

$$[s^2 Y(s) - s y(0^-) - y'(0^-)] + 7[s Y(s) - y(0^-)] + 10 Y(s) = 2[s X(s) - x(0^-)] + X(s)$$

$$[s^2 Y(s) - s - 1] + 7[s Y(s) - 1] + 10 Y(s) = 2[s X(s) - 1] + X(s)$$

$$\Rightarrow Y(s) = \frac{(2s+1)X(s) + s+6}{s^2+7s+10}$$

The natural response is associated with the modes only

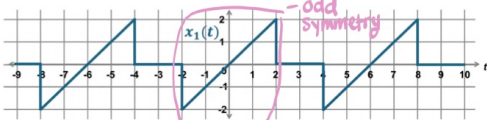
$$Y(s) = \frac{(2s+1) \cdot \frac{1}{s+5} + s+6}{s^2+7s+10} = \frac{1}{3(s+2)} + \frac{5}{2(s+3)} - \frac{11}{6(s+5)}$$

$$Y_{nat}(s) = \frac{1}{3(s+2)} - \frac{11}{6(s+5)} \quad \text{--- (B)}$$



5. (25 points) Fourier Series and Signals

- a. (8 points) Find the Fourier Series of the following signal $x_1(t)$ in cosine/sine form



S1: Find T_0 $T_0 = -(-4) = 6$

S2: Find ω_0 $\omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{6} = \frac{\pi}{3}$

S3: Determine symmetry

odd
 $a_0 = 0, a_n = 0$

S4: Get b_n using formula

$$b_n = \frac{2}{T_0} \int_{-T_0/2}^{T_0/2} x(t) \sin(n\omega_0 t) dt$$

$$= \frac{1}{3} \int_{-2}^2 t \sin(n\omega_0 t) dt \leftarrow \text{Up to here}$$

Integration by parts: $\downarrow$ Incorrect but for 1 pt; I'm good

$$u = t \quad dv = \sin(n\omega_0 t) \quad du = dt \quad v = -\frac{\cos(n\omega_0 t)}{n\omega_0}$$

$$= t \left(-\frac{\cos(n\omega_0 t)}{n\omega_0} \right) - \int -\frac{\cos(n\omega_0 t)}{n\omega_0} dt$$

$$= -\frac{t \cos(n\omega_0 t)}{n\omega_0} + \frac{1}{n\omega_0} \int \cos(n\omega_0 t) dt$$

$$= -\frac{t \cos(n\omega_0 t)}{n\omega_0} + \frac{1}{n\omega_0} \left(\frac{\sin(n\omega_0 t)}{n\omega_0} \right)$$

$$= -\frac{t \cos(n\omega_0 t)}{n\omega_0} + \frac{\sin(n\omega_0 t)}{(n\omega_0)^2}$$

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- d. (6 points) Given that the Fourier transform of $g(t) = e^{-\frac{t^2}{2}}$ is $G(\omega) = e^{-\frac{\omega^2}{2}}$, find the Fourier transform of $x_4(t) = e^{-\frac{(t+1)^2}{2}} + 2e^{-\frac{t^2}{2}} + e^{-\frac{(t-1)^2}{2}}$. What kind of filter does $x_4(t)$ represent and why?

$$e^{-\frac{t^2}{2}} \leftrightarrow e^{-\frac{\omega^2}{2}}$$

Using the time shifting property

$$e^{-\frac{(t+1)^2}{2}} \leftrightarrow e^{-\frac{\omega^2}{2}} e^{-j\omega} \quad \text{(A)}$$

$$e^{-\frac{(t-1)^2}{2}} \leftrightarrow e^{-\frac{\omega^2}{2}} e^{j\omega} \quad \text{(B)}$$

$$x_4(t) \leftrightarrow e^{-\frac{\omega^2}{2}} e^{j\omega} + 2e^{-\frac{\omega^2}{2}} + e^{-\frac{\omega^2}{2}} e^{-j\omega} \quad \text{(C)}$$

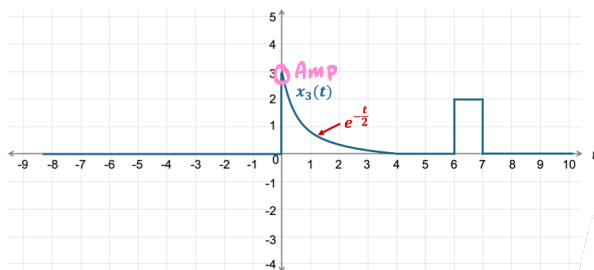
As $\omega \rightarrow 0$, $e^{-\frac{\omega^2}{2}} \rightarrow 1$ and as $\omega \rightarrow \infty$, $e^{-\frac{\omega^2}{2}} \rightarrow 0$. (D)

Each of the three terms corresponds to a low-pass filter, and the combination is also a low-pass filter.

(E)

3. (10 points) Laplace Transform & Its Inverse

- a. (5 points) Determine the Laplace transform $X_3(s)$ of the waveform:



Laplace domain

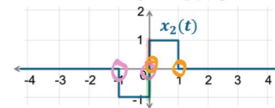
$$x_3(t) = 3e^{-\frac{t}{2}}u(t) + 2u(t-6) - 2u(t-7)$$

$$X_3(s) = \frac{3}{s + \frac{1}{2}} + \frac{2}{s} e^{-6s} - \frac{2}{s} e^{-7s}$$

The initial value $x_3(0+) = 3$ (from graph above) (A)

And final value $x_3(\infty) = 0$ (from graph above) (B)

- b. (5 points) Find the Fourier Transform of the signal $x_2(t)$ plotted below.



Using the definition of the Fourier Transform:

$$\begin{aligned} X_2(\omega) &= \int_{-\infty}^{\infty} x_2(t) e^{-j\omega t} dt = \int_{-1}^1 1 e^{-j\omega t} dt + \int_1^2 1 e^{-j\omega t} dt \\ &= -\int_{-1}^0 e^{-j\omega t} dt + \int_0^1 e^{-j\omega t} dt = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-1}^0 + \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^1 \\ &= \frac{1 - e^{j\omega}}{j\omega} + \frac{1 - e^{-j\omega}}{j\omega} = \frac{2 - e^{j\omega} - e^{-j\omega}}{j\omega} \quad \text{(C)} \\ &= \frac{2 - 2 \cos \omega}{j\omega} = \frac{4 \sin^2(\frac{\omega}{2})}{j\omega} \quad \text{(D)} \end{aligned}$$

1 point for trying. 1 each for A-D.

- c. (6 points) The Fourier transform of a signal $x(t)$ is $X(\omega)$. Find the Fourier transform of the signal

$$y(t) = x(2t-4) * x\left(\frac{t-2}{3}\right), \text{ where } * \text{ represents convolution.}$$

Work Cont.

$$x(2t-4) \Rightarrow x(2(t-2))$$

$$\Rightarrow x(2t) \cdot x(t-2) \Rightarrow \frac{1}{2} X\left(\frac{\omega}{2}\right) e^{-j\omega 2}$$

$$x\left(\frac{t-2}{3}\right) \Rightarrow x\left(\frac{1}{3}(t-2)\right)$$

$$\Rightarrow x\left(\frac{1}{3}t\right) \cdot x\left(\frac{t-2}{3}\right) \Rightarrow \frac{1}{3} X\left(\frac{\omega}{3}\right) e^{-j\omega 2}$$

$$\Rightarrow 3X(3\omega) e^{-j\omega 2}$$

Up until here 4 pts

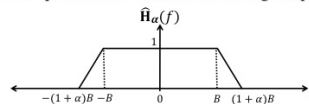
E 3337: Signals and Systems Analysis

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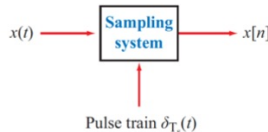
Q 6c

- c. (10 points) Audio signal frequencies range from 20Hz to 20kHz, and we wish to digitize stereophonic sound using the following available components:

- Low-pass filters with the following frequency response ($\alpha > 0$, B are adjustable):



- A signal sampling system with a maximum sampling rate of 44kHz.



- a. (5 points) Calculate the optimal values of B and α to achieve signal digitization to ensure that the signal for a single channel can be recovered without any distortion.

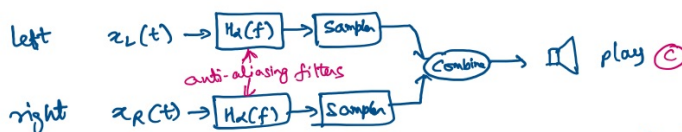
For each channel:



Grading: 1 pt for trying + 2 pts each for (A) & (B)

$$\begin{aligned} f_s &= 2B(1+\alpha) = 44 \text{ kHz} \\ B &= 20 \text{ kHz} \quad \text{(A)} \\ \Rightarrow 1+\alpha &= \frac{44}{40} = 1.1 \\ \Rightarrow \alpha &= 0.1 \quad \text{(B)} \end{aligned}$$

- a. (5 points) Sketch a diagram showing how these components and their parameters must be chosen and interconnected to build a stereophonic (left and right) sound digitizing system that avoids aliasing.



2. (10 points) Linear Time-Invariant Systems

- a. (5 points) An LTI system has the impulse response $h(t) = \cos(t)u(t)$. Calculate the response to the input signal $x(t) = \cos(t)u(t)$ and determine whether or not this system is BIBO stable.

Time-domain solution: $y(t) = \cos(t)u(t) * \cos(t)u(t)$ (A)

$$= (e^{jt} + e^{-jt})u(t) * (e^{jt} + e^{-jt})u(t)$$

$$= \frac{1}{4} \{ e^{jt}u(t) * e^{jt}u(t) + e^{jt}u(t) * e^{-jt}u(t) + e^{-jt}u(t) * e^{jt}u(t) + e^{-jt}u(t) * e^{-jt}u(t) \}$$

$$= \frac{1}{4} \{ t e^{j2t} + 2 \left(\frac{e^{jt} - e^{-jt}}{2j} \right) + t e^{-j2t} \} u(t) = \frac{1}{2} t \cos(t) u(t) + \frac{1}{2} \sin(t) u(t)$$

unbounded $\Rightarrow$ NOT BIBO stable

Frequency-domain solution:

$$Y(s) = \left(\frac{s}{s^2+1} \right) \left(\frac{s}{s^2+1} \right) = \frac{s^2}{(s^2+1)^2}$$

Laplace $X(s)H(s)$

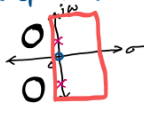
$$(s^2+1)^2 = 0$$

$$s^2+1=0$$

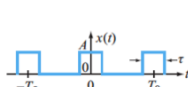
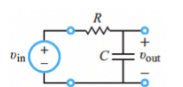
$$s^2 = -1$$

$$s = \pm j \Rightarrow \text{imaginary pole}$$

NOT BIBO stable since the poles are on the j axis, so NOT in the open left-half plane.



- b. (5 points) Calculate the response of the following system to the input $x(t)$.



$$x(t) = \frac{A\tau}{T_0} + \sum_{n=1}^{\infty} \frac{2A}{n\pi} \sin\left(\frac{n\pi\tau}{T_0}\right) \cos\left(\frac{2n\pi t}{T_0}\right)$$

$$H(\omega) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{1}{1+j\omega RC}$$

(A) $|H(\omega)| = \frac{1}{\sqrt{1+(\omega RC)^2}}$, $\angle H(\omega) = -\tan^{-1}(\omega RC)$

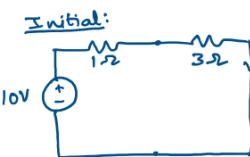
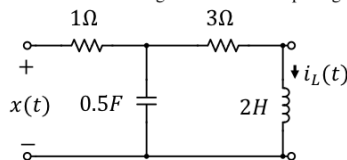
$$\Rightarrow y(t) = \frac{A\tau}{T_0} |H(0)| + \sum_{n=1}^{\infty} \frac{2A}{n\pi} |H(n\omega_0)| \sin\left(\frac{n\pi\tau}{T_0}\right) \cos\left(\frac{2n\pi t}{T_0} - \angle H(n\omega_0)\right)$$

$$= \frac{A\tau}{T_0} + \sum_{n=1}^{\infty} \frac{2A}{n\pi \sqrt{1+(n\omega_0 RC)^2}} \sin\left(\frac{n\pi\tau}{T_0}\right) \cos\left(\frac{2n\pi t}{T_0} - \tan^{-1}(n\omega_0 RC)\right)$$

where $\omega_0 = \frac{2\pi}{T_0}$

4. (20 points) Applications of the Laplace Transform

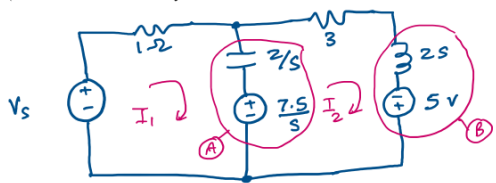
Consider the following circuit and the input signal $x(t) = 10 - 5u(t)$ V.



$$i_L(0^-) = 2.5 \text{ A}$$

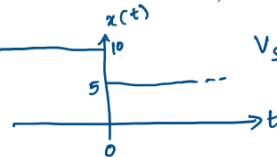
$$v_C(0^-) = 7.5 \text{ V}$$

- a. (2 points) Sketch the Laplace-domain version of the given circuit with the initial conditions indicated (use the back sheet for your work as needed and write the final answer below).



Grading: 1 pt each (A), (B)

- b. (3 points) Write the Laplace transform of the input for $t \geq 0$ (use the back sheet for your work and write the final answer below).



$$V_s = \frac{5}{s} \text{ for } t \geq 0 \text{ part.}$$

Grading: 1 pt trying, 3 pts correct answer

- c. (5 points) Calculate the Laplace-domain current $I_L(s)$ for $t \geq 0$. (use the back sheet for your work and write the final answer below). Your answer should be in the form of a ratio of two polynomials.

$$\text{mesh equations: } \frac{5}{s} - \frac{7.5}{s} = I_1 \left(1 + \frac{2}{s}\right) - I_2 \left(\frac{2}{s}\right)$$

$$\frac{7.5}{s} + 5 = -I_1 \left(\frac{2}{s}\right) + I_2 \left(\frac{2}{s} + 3 + 2s\right)$$

$$\Delta = \begin{vmatrix} 1 + \frac{2}{s} & -\frac{2}{s} \\ -\frac{2}{s} & 2s + 3 + \frac{2}{s} \end{vmatrix} = \frac{2s^2 + 7s + 8}{s}$$

$$\Delta_2 = \begin{vmatrix} 1 + \frac{2}{s} & -\frac{2.5}{s} \\ -\frac{2}{s} & 5 + \frac{7.5}{s} \end{vmatrix} = \frac{5s^2 + 17.5s + 10}{s^2}$$

$$\Rightarrow I_L(s) = \frac{\Delta_2}{\Delta} = \frac{5s^2 + 17.5s + 10}{s(2s^2 + 7s + 8)}$$

- d. An LTI system has the following differential equation description, with initial conditions $x(0^-) = 1$, $y(0^-) = 1$, $x'(0^-) = 1$, $y'(0^-) = 1$, $y''(0^-) = 1$, and input $x(t) = e^{-2t}u(t)$:

$$\frac{d^2y}{dt^2} + 8 \frac{dy}{dt} + 15y = \frac{dx}{dt} + 1x \Rightarrow H(s) = \frac{s+4}{(s+3)(s+5)}$$

$$X(s) = \frac{1}{(s+2)}$$

- (i) (5 points) Identify the poles, zeros, and modes of this system.

$$s^2 + 8s + 15 = (s+3)(s+5)$$

$$\text{poles: } \{-3, -5\}$$

$$\text{zeros: } \{-4\}$$

$$\text{modes: } \{-3, -5\}$$

Grading: 1 pt trying, 2 pts (A), 1 pt (B), 1 pt (C)

- (ii) (5 points) Calculate the total response of this system in the Laplace domain. Your answer should be expressed as a ratio of polynomials.

Transform each term of the LCCDE:

$$[s^2 Y(s) - s y(0^-) - y'(0^-)] + 8[s Y(s) - y(0^-)] + 15 Y(s) = s X(s) - x(0^-) + 4 X(s)$$

$$[s^2 Y(s) - s - 1] + 8[s Y(s) - 1] + 15 Y(s) = \frac{s}{s+2} - 1 + \frac{4}{(s+2)}$$

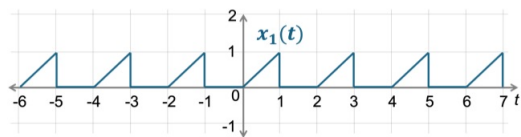
$$(s^2 + 8s + 15) Y(s) = \frac{s+4}{s+2} + s + 8 = \frac{s^2 + 11s + 20}{(s+2)}$$

$$\Rightarrow Y(s) = \frac{s^2 + 11s + 20}{(s+2)(s+3)(s+5)} = \frac{s^2 + 11s + 20}{(s^3 + 10s^2 + 31s + 30)}$$

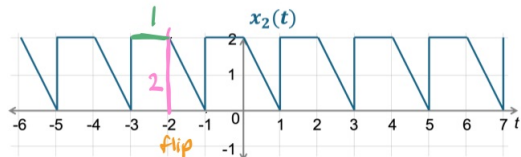
5. (25 points) Fourier Series and Transform

a. (7 points). The Fourier Series of the following signal $x_1(t)$ is

$$x_1(t) = \frac{1}{4} + \sum_{n=1}^{\infty} \left\{ \frac{1}{n^2 \pi^2} [\cos(n\pi) - 1] \cos(n\pi t) - \frac{1}{n\pi} \cos(n\pi) \sin(n\pi t) \right\}$$



Find the Fourier Series of the signal $x_2(t)$ graphed below in cosine/sine form.



2 points for trying.

Observe that $x_2(t)$ is a transformed version of $x_1(t)$. Specifically

$$x_2(t) = 2(1 - x_1(t)) = 2 - 2x_1(t)$$

Substituting $x_1(t)$, $x_2(t) = 2 - 2\frac{1}{4} - \sum_{n=1}^{\infty} \left\{ \frac{2}{n^2 \pi^2} [\cos(n\pi) - 1] \cos(n\pi t) - \frac{2}{n\pi} \cos(n\pi) \sin(n\pi t) \right\}$

$$x_2(t) = \frac{3}{2} - \sum_{n=1}^{\infty} \left\{ \frac{2}{n^2 \pi^2} [\cos(n\pi) - 1] \cos(n\pi t) - \frac{2}{n\pi} \cos(n\pi) \sin(n\pi t) \right\}$$

b. (6 points) The Fourier series of a signal $x_3(t)$ in complex exponential notation is

$$x_3(t) = \sum_{n=-\infty}^{\infty} e^{-jn3\omega_0} \cos n\pi\omega_0 e^{jn\omega_0 t}$$

Find the Fourier Series of $x_3(t)$ in cosine/sine form. Please write your final answer below:

$$a_0 = 1 \rightarrow e^{-j(0)3\omega_0} \cos(0) \pi\omega_0 = e^{j2\omega_0} \omega_0 t = 1$$

$$a_n = e^{-jn3\omega_0} \cos n\pi\omega_0 = e^{-jn3\omega_0} [\cos(n\omega_0 t) + j \sin(n\omega_0 t)]$$

$$b_n = j e^{-jn3\omega_0} \cos n\pi\omega_0$$

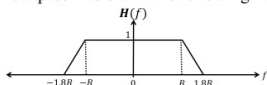
$$x_3(t) = \sum_{n=-\infty}^{\infty} e^{-jn3\omega_0} \cos n\pi\omega_0 e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} e^{-jn3\omega_0} \cos n\pi\omega_0 [\cos(n\omega_0 t) + j \sin(n\omega_0 t)]$$

$$= \sum_{n=-\infty}^{\infty} e^{-jn3\omega_0} \cos n\pi\omega_0 \cos(n\omega_0 t) + j \sum_{n=-\infty}^{\infty} e^{-jn3\omega_0} \cos n\pi\omega_0 \sin(n\omega_0 t)$$

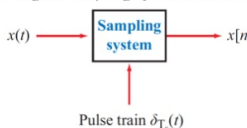
For $n=0$, $x_0 = a_0 = e^{j0} \times \cos(0) = 1 \times 1 = 1$

c. (10 points) Audio signal frequencies range from 20Hz to 20kHz, and we wish to digitize them using the following available components:

- Low-pass filters with the following frequency response (B is adjustable):

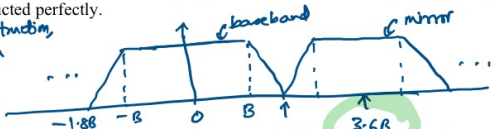


- A signal sampling system with an adjustable sampling rate.



Sketch a diagram showing how these components will be adjusted and interconnected. Find the value of B and the minimum sampling rate that will allow the signal to be sampled and reconstructed perfectly.

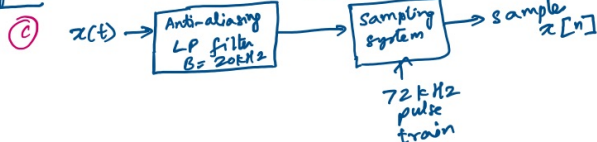
For perfect reconstruction, the mirror spectra should not overlap with the baseband spectrum



$$\Rightarrow B = 20 \text{ kHz} \quad \text{setting for LP filter}$$

$$\text{Sampling rate} = 3.6 \times 20 = 72 \text{ kHz} \quad \text{setting for sampling system}$$

Diagram:



c. (6 points) The Fourier transform of a signal $x(t)$ is $X(\omega)$

$$y(t) = x\left(\frac{t}{2}\right) * x\left(\frac{t-3}{2}\right), \text{ where } * \text{ represents convolution}$$

$$x\left(\frac{t}{2}\right) = 2X(2\omega), \quad x\left(\frac{t-3}{2}\right) = 2X(2\omega) e^{-j3\omega}$$

$$x_1(t) * x_2(t) = X_1(\omega) \cdot X_2(\omega) = 4[X(2\omega)]^2 e^{-j3\omega}$$

d. (6 points) Find the Fourier transform of $x_4(t) = 2 \text{sinc}(t + 2\pi) + 3 \text{sinc}(t) + 4 \text{sinc}(t - 2\pi)$. What kind of filter does $x_4(t)$ represent and why?

Hint: $\text{sinc}(t) \leftrightarrow \pi \text{rect}\left(\frac{\omega}{2}\right)$. 1 point for trying.

$$\text{sinc}(t) \leftrightarrow \pi \text{rect}\left(\frac{\omega}{2}\right)$$

$$3 \text{sinc}(t) \leftrightarrow 3\pi \text{rect}\left(\frac{\omega}{2}\right)$$

$$2 \text{sinc}(t + 2\pi) \leftrightarrow 2\pi \text{rect}\left(\frac{\omega}{2}\right) e^{j2\pi\omega} \quad (\text{Table 5-7, \#4})$$

$$4 \text{sinc}(t - 2\pi) \leftrightarrow 4\pi \text{rect}\left(\frac{\omega}{2}\right) e^{-j2\pi\omega} \quad (\text{Table 5-7, \#4})$$

$$X_4(\omega) = 2\pi \text{rect}\left(\frac{\omega}{2}\right) e^{j2\pi\omega} + 3\pi \text{rect}\left(\frac{\omega}{2}\right) + 4\pi \text{rect}\left(\frac{\omega}{2}\right) e^{-j2\pi\omega}$$

$$= \pi \text{rect}\left(\frac{\omega}{2}\right) [2e^{j2\pi\omega} + 3 + 4e^{-j2\pi\omega}]$$

Multiplication with $\text{rect}\left(\frac{\omega}{2}\right)$ implies that $X_4(\omega) = 0$ for $|\omega| > 1$. Therefore, $x_4(t)$ is a low pass filter.

b. (20 points) Applications of Fourier methods

a. (5 points) What type of filter does the following system represent in the frequency domain when $L = 0.01 \text{H}$, $C = 0.01 \text{F}$, and $R = 0.1 \Omega$?

$$LC \frac{d^2 y(t)}{dt^2} + RC \frac{dy(t)}{dt} + y(t) = x(t) + LC \frac{d^2 x(t)}{dt^2}$$

$$H(\omega) = \frac{1 + (j\omega)^2 LC}{(j\omega)^2 LC + (j\omega)RC + 1}$$

$$= 0 \text{ when } \omega^2 LC = 1, \text{ at } \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\text{when } L = 0.01 \text{ H}, C = 0.01 \text{ F}, LC = 10^{-4}$$

$$\Rightarrow \omega_0 = 100 \text{ rad/sec}$$

This is a band-stop filter @ 100 rad/sec

Calculate the center frequency, bandwidth, and quality factor Q for the filter in part (a) above.

$$\omega_0 = 100 \text{ rad/sec}$$

$$B = \frac{R}{L} = \frac{0.1}{0.01} = 10 \text{ rad/sec}$$

$$Q = \frac{\omega_0}{B} = \frac{100}{10} = 10$$

2. (10 points) Linear Time-Invariant Systems

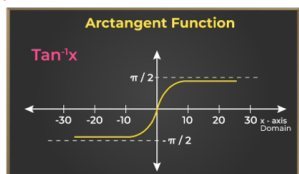
- a. (5 points) An LTI system has the impulse response $h(t) = e^{-t} \cos(t) u(t)$. Is this system BIBO stable? Calculate the response $y(t)$ to the input signal $x(t) = t u(t)$.

$$H(s) = \frac{s+1}{(s+1)^2 + 1}, \quad X(s) = \frac{1}{s^2} \Rightarrow Y(s) = \frac{(s+1)}{(s+1)^2 + 1} \cdot \frac{1}{s^2}$$

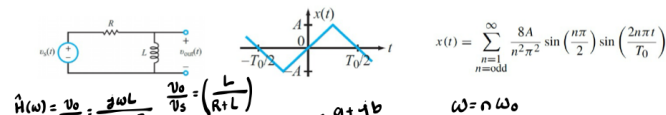
poles: $\{-1 \pm j\}$

$$\Rightarrow y(t) = \frac{1}{2} u(t) - \frac{1}{2} e^{-t} \sin(t) u(t)$$

BIBO STABLE



- b. (5 points) Calculate the response $v_{out}(t)$ of the following system to the input $v_s(t) = x(t)$.



$$\hat{H}(\omega) = \frac{v_o}{v_s} = \frac{j\omega L}{j\omega L + R}$$

$$|H(\omega)| = \frac{\omega L}{\sqrt{\omega^2 L^2 + R^2}}$$

$$\angle H(\omega) = \tan^{-1}\left(\frac{\omega L}{R}\right) - \tan^{-1}\left(\frac{\omega L}{R}\right)$$

$$= \frac{\pi}{2} - \tan^{-1}\left(\frac{\omega L}{R}\right)$$

Plug in:

$$\sum_{n=1}^{\infty} \frac{8A}{n^2 \pi^2} \left(\frac{n\omega_0 L}{\sqrt{R^2 + n^2 \omega_0^2 L^2}} \right) \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{2n\pi t}{T_0} + \frac{\pi}{2} - \tan^{-1}\left(\frac{n\omega_0 L}{R}\right)\right)$$

- b. (6 points) The signal $x_2(t)$ is defined as follows

$$x_2(t) = \sum_{n=0}^{\infty} e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0) e^{jn\omega_0 t}$$

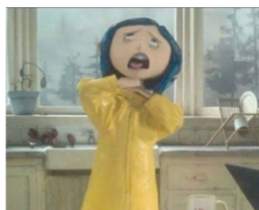
Find the Fourier Series of $x_2(t)$ in cosine/sine form. Please write your final answer below:

$$a_0 = 0$$

$$a_n = \frac{e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0)}{j}$$

$$b_n = \frac{j e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0)}{1}$$

$$e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0) \cos(n\omega_0 t) + e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0) (j \sin(n\omega_0 t))$$



3. (10 points) Laplace Transform & Its Inverse

- a. (5 points) Derive the Laplace transform of $x(t) = A e^{-t} \sin(\omega_0 t + \theta) u(t)$.

What are the initial and final values of this signal?

$$\sin(\omega_0 t) \cos(\theta) + \cos(\omega_0 t) \sin(\theta)$$

$$A e^{-t} \sin(\omega_0 t) \cos(\theta) + \cos(\omega_0 t) \sin(\theta) u(t)$$

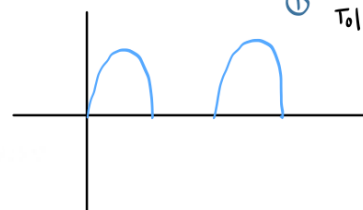
$$A \left(\frac{\omega_0}{s^2 + \omega_0^2} \cos(\theta) + \frac{s}{s^2 + \omega_0^2} \sin(\theta) \right)$$

initial value: $A \sin(\theta)$

final value: 0

5. (25 points) Fourier Series and Transform

- a. (7 points) Find the Fourier Series of the signal $x(t) = |\sin(t)|$ in complex exponential representation where $||$ indicates the absolute value.



$$e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

$$e^{-j\theta} = \cos(\theta) - j \sin(\theta)$$

Trigonometric Relationship

$$e^{j\theta} + e^{-j\theta} = 2 \cdot \cos(\theta)$$

$$e^{j\theta} - e^{-j\theta} = 2j \cdot \sin(\theta)$$

$$\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$\omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{\pi} = 2$$

$$|x_n| = \frac{C_n}{2}$$

$$x_n = \frac{1}{\pi} \int_0^\pi \sin(t) e^{-jn2t} dt$$

$$\begin{aligned} x_n &= \frac{1}{\pi} \int_0^\pi \sin(t) e^{-jn2t} dt \\ &= \frac{1}{\pi} \int_0^\pi \left(\frac{e^{jt} - e^{-jt}}{2j} \right) e^{-jn2t} dt \\ &= \frac{1}{2j\pi} \int_0^\pi (e^{jt} - e^{-jt}) e^{-jn2t} dt \\ &= \frac{1}{2j\pi} \int_0^\pi (e^{jt(1-2n)} - e^{-jt(1+2n)}) dt \\ &= \frac{1}{2j\pi} \left[\frac{e^{jt(1-2n)}}{(1-2n)} - \frac{e^{-jt(1+2n)}}{(1+2n)} \right]_0^\pi \\ &= \frac{1}{2j\pi} \left(\frac{e^{j\pi(1-2n)}}{(1-2n)} - \frac{e^{-j\pi(1+2n)}}{(1+2n)} - \left(\frac{1}{(1-2n)} - \frac{1}{(1+2n)} \right) \right) \end{aligned}$$

$$\Rightarrow \frac{1}{2j\pi} \left(\frac{e^{j(1-2n)\pi}}{(1-2n)} - \frac{e^{-j(1+2n)\pi}}{(1+2n)} \right)$$

$$= \frac{1}{2j\pi} \left(\frac{e^{j(1-2n)\pi} - 1}{(1-2n)} - \frac{e^{-j(1+2n)\pi} - 1}{(1+2n)} \right)$$

$$= \frac{1}{2j\pi} \left(\frac{e^{j(1-2n)\pi} - 1}{(1-2n)} + \frac{e^{-j(1+2n)\pi} - 1}{(1+2n)} \right)$$

$e^{j(1-2n)\pi}$ and $e^{-j(1+2n)\pi}$ are always -1 (check by substituting $n = 1, 2, 3, 4 \dots$). Therefore,

$$x_n = \frac{-1}{2\pi} \left(\frac{-2}{(1-2n)} + \frac{-2}{(1+2n)} \right)$$

$$x_n = \frac{2}{2\pi} \left(\frac{1}{(1-2n)} + \frac{1}{(1+2n)} \right)$$

$$x_n = \frac{1}{\pi} \left(\frac{2}{(1-4n^2)} \right)$$

$$x_n = \frac{2}{\pi} \frac{1}{(1-4n^2)}$$



- c. (6 points) The Fourier transform of a signal $x(t)$ is $X(\omega)$. Find the Fourier transform of the signal

$$v(t) = x_f(2t) * x_f(-3t-3), \text{ where } * \text{ represents convolution}$$

$$x_1(2t) \Rightarrow \frac{1}{2} X\left(\frac{\omega}{2}\right)$$

$$x_2(-3t-3) \Rightarrow$$

$$x_2(-3t) = \frac{1}{3} X\left(-\frac{\omega}{3}\right)$$

$$x_2[3(t-1)] = e^{-j\omega(-1)} X(\omega) = e^{j\omega} X(\omega)$$

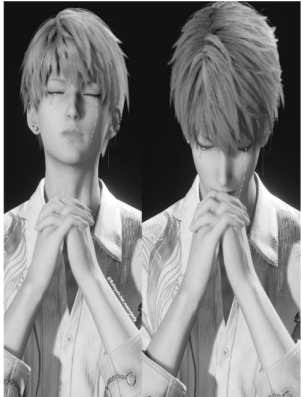
$$= e^{j\omega} \left(\frac{1}{3} \right) X\left(-\frac{\omega}{3}\right)$$



d. (6 points) Find the Fourier transform of $x_4(t) = \text{sinc}(t - 10)e^{j100t}$. What kind of filter does $x_4(t)$

represent and why?

Hint: $\text{sinc}(t) \xrightarrow{\mathcal{F}} \pi \text{rect}\left(\frac{\omega}{2}\right)$



$$\text{sinc}(t) - \text{sinc}(10)$$

$$\pi \text{rect}\left(\frac{\omega}{2}\right) = g(\omega)$$

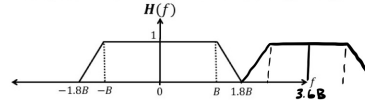
$$g(t-10) \Rightarrow e^{-j\omega 10}$$

$$g(\omega)e^{-j\omega 10}$$

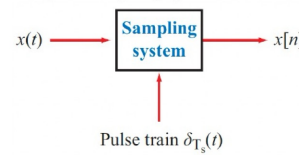
$$\left(\pi \text{rect}\left(\frac{\omega}{2}\right)e^{j\omega 10}\right)e^{j100t}$$

c. (10 points) If the signal $x(t)$ is bandlimited to 10kHz, calculate the minimum sampling rate for the signal $y(t) = x^2(t)$ using the following available components that will ensure the sampled signal can be reconstructed perfectly.

- Low-pass filters with the following frequency response (B is adjustable):



- A signal sampling system with an adjustable sampling rate $f_s = 1/T_s$.

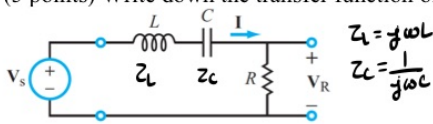


$$B = 20 \text{ kHz}$$

Sketch a diagram showing how these components will be adjusted and interconnected.

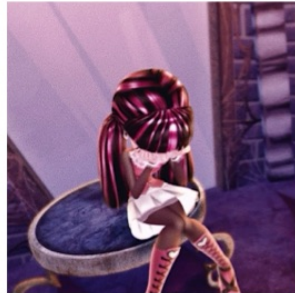
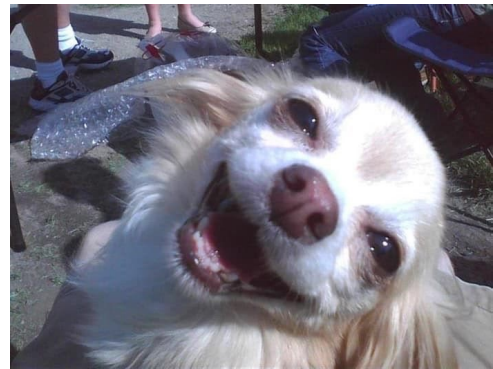
6. (20 points) Applications of Fourier Methods

- a. (5 points) Write down the transfer function of the following circuit in the frequency domain.



$$V_R = V_s \left(\frac{R}{Z_L + Z_C + R} \right)$$

$$\hat{H}(\omega) = \frac{V_R}{V_s} = \frac{R}{\frac{1}{j\omega C} + j\omega L + R} \left(\frac{j\omega C}{j\omega C} \right) = \frac{j\omega RC}{1 + j\omega^2 LC + j\omega RC} = \frac{j\omega RC}{(1 - \omega^2 LC) + j\omega RC}$$



- b. (5 points) Calculate the center frequency ω_0 , bandwidth B , and the frequency ω_{45} at which the phase shift of the filter in part (a) above is 45° , when $L = 2H$, $C = 1F$, and $R = 1\Omega$.

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2}} \text{ [rad/sec]}$$

$$B = \frac{R}{L} = \frac{1}{2} \text{ [rad/sec]}$$

$$\phi(\omega) = \angle H(\omega) = \angle N(s) - \angle D(s) = \frac{\pi}{2} - \tan^{-1}\left(\frac{j\omega RC}{1 - \omega^2 LC}\right)$$

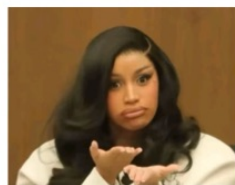
$$\frac{\pi}{2} - \tan^{-1}\left(\frac{j\omega RC}{1 - \omega^2 LC}\right) = \frac{\pi}{4} \Rightarrow \frac{j\omega RC}{1 - \omega^2 LC} = 1$$

$$\frac{j\omega}{1 - 2\omega^2} = 1 \Rightarrow \frac{\omega}{1 - 2\omega^2} = 1$$

$$1 - 2\omega^2 = \omega \Rightarrow 2\omega^2 - \omega + 1 = 0$$

$$\angle N(s) = \tan^{-1}\left(\frac{\text{imag}}{\text{real}}\right) = \tan^{-1}\left(\frac{j\omega RC}{0}\right) = \tan^{-1}(\infty) = \frac{\pi}{2}$$

$$\angle D(s) = \tan^{-1}\left(\frac{\text{imag}}{\text{real}}\right) = \tan^{-1}\left(\frac{j\omega RC}{1 - \omega^2 LC}\right)$$



$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(2)(1)}}{2(-2)} = \frac{1 \pm \sqrt{1 - 8}}{-4} = \frac{1 \pm \sqrt{-7}}{-4} = \frac{1 \pm j\sqrt{7}}{-4}$$

Laplace Transform Pairs

	$x(t)$	$\mathbf{X}(s) = \mathcal{L}[x(t)]$
1	$\delta(t)$	1
1a	$\delta(t-T)$	e^{-Ts}
2	$u(t)$	$\frac{1}{s}$
2a	$u(t-T)$	$\frac{e^{-Ts}}{s}$
3	$e^{-at} u(t)$	$\frac{1}{s+a}$
3a	$e^{-a(t-T)} u(t-T)$	$\frac{e^{-Ts}}{s+a}$
4	$t u(t)$	$\frac{1}{s^2}$
4a	$(t-T) u(t-T)$	$\frac{e^{-Ts}}{s^2}$
5	$t^2 u(t)$	$\frac{2}{s^3}$
6	$te^{-at} u(t)$	$\frac{1}{(s+a)^2}$
7	$t^2 e^{-at} u(t)$	$\frac{2}{(s+a)^3}$
8	$t^{n-1} e^{-at} u(t)$	$\frac{(n-1)!}{(s+a)^n}$
9	$\sin(\omega_0 t) u(t)$	$\frac{\omega_0}{s^2 + \omega_0^2}$
10	$\sin(\omega_0 t + \theta) u(t)$	$\frac{s \sin \theta + \omega_0 \cos \theta}{s^2 + \omega_0^2}$
11	$\cos(\omega_0 t) u(t)$	$\frac{s}{s^2 + \omega_0^2}$
12	$\cos(\omega_0 t + \theta) u(t)$	$\frac{s \cos \theta - \omega_0 \sin \theta}{s^2 + \omega_0^2}$
13	$e^{-at} \sin(\omega_0 t) u(t)$	$\frac{\omega_0}{(s+a)^2 + \omega_0^2}$
14	$e^{-at} \cos(\omega_0 t) u(t)$	$\frac{s+a}{(s+a)^2 + \omega_0^2}$
15	$2e^{-at} \cos(bt - \theta) u(t)$	$\frac{e^{-j\theta}}{s+a+jb} + \frac{e^{-j\theta}}{s+a-jb}$
15a	$e^{-at} \cos(bt - \theta) u(t)$	$\frac{(s+a)^2 + b^2}{(s+a+jb)^n + (s+a-jb)^n}$
16	$\frac{2t^{n-1}}{(n-1)!} e^{-at} \cos(bt - \theta) u(t)$	$\frac{e^{j\theta}}{(s+a+jb)^n} + \frac{e^{-j\theta}}{(s+a-jb)^n}$

Figure 5.4.2: Fourier series expressions for a select set of periodic waveforms.

Waveform	Fourier Series
1. Square Wave	$x(t) = \sum_{n=1}^{\infty} \frac{4A}{n\pi} \sin\left(\frac{n\pi}{2}\right) \cos\left(\frac{2n\pi t}{T_0}\right)$
2. Time-Shifted Square Wave	$x(t) = \sum_{n=1}^{\infty} \frac{4A}{n\pi} \sin\left(\frac{2n\pi t}{T_0}\right)$
3. Pulse Train	$x(t) = \frac{A\tau}{T_0} + \sum_{n=1}^{\infty} \frac{2A}{n\pi} \sin\left(\frac{n\pi\tau}{T_0}\right) \cos\left(\frac{2n\pi t}{T_0}\right)$
4. Triangular Wave	$x(t) = \sum_{n=1}^{\infty} \frac{8A}{n^2\pi^2} \cos\left(\frac{2n\pi t}{T_0}\right)$
5. Shifted Triangular Wave	$x(t) = \sum_{n=1}^{\infty} \frac{8A}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \cos\left(\frac{2n\pi t}{T_0}\right)$
6. Sawtooth	$x(t) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2A}{n\pi} \sin\left(\frac{2n\pi t}{T_0}\right)$
7. Backward Sawtooth	$x(t) = \frac{A}{2} + \sum_{n=1}^{\infty} \frac{A}{n\pi} \sin\left(\frac{2n\pi t}{T_0}\right)$
8. Full-Wave Rectified Sinusoid	$x(t) = \frac{2A}{\pi} + \sum_{n=1}^{\infty} \frac{4A}{\pi(1-4n^2)} \cos\left(\frac{2n\pi t}{T_0}\right)$
9. Half-Wave Rectified Sinusoid	$x(t) = \frac{A}{\pi} + \frac{A}{2} \sin\left(\frac{2n\pi t}{T_0}\right) + \sum_{n=2,4,6,\dots}^{\infty} \frac{2A}{\pi(1-n^2)} \cos\left(\frac{2n\pi t}{T_0}\right)$

$\sin x = \pm \cos(x \mp 90^\circ)$	$\sin x + \sin y = 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right)$
$\cos x = \pm \sin(x \pm 90^\circ)$	$\sin x - \sin y = 2 \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right)$
$\sin x = -\sin(x \pm 180^\circ)$	$\cos x + \cos y = 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right)$
$\cos x = -\cos(x \pm 180^\circ)$	$\cos x - \cos y = -2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right)$
$\sin(-x) = -\sin x$	$e^{jx} = \cos x + j \sin x$
$\cos(-x) = \cos x$	$\sin x = \frac{e^{jx} - e^{-jx}}{2j}$
$\sin^2 x = \frac{1}{2} (1 - \cos 2x)$	$\cos x = \frac{e^{jx} + e^{-jx}}{2}$
$\cos^2 x = \frac{1}{2} (1 + \cos 2x)$	
$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$	
$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$	
$2 \sin x \sin y = \cos(x-y) - \cos(x+y)$	
$2 \sin x \cos y = \sin(x+y) + \sin(x-y)$	
$2 \cos x \cos y = \cos(x+y) + \cos(x-y)$	
$\sin 2x = 2 \sin x \cos x$	
$\cos 2x = 1 - 2 \sin^2 x$	

Most of these formulas can be derived from Euler's identity, for example:

$$e^{j(A-B)} = \cos(A-B) + j \sin(A-B)$$

$$e^{jA} e^{-jB} = (\cos A + j \sin A) (\cos B - j \sin B)$$

$$= (\cos A \cos B + \sin A \sin B) + j(\sin A \cos B - \cos A \sin B)$$

Comparing the real and imaginary terms, we get:

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

Table 3-1: Properties of the Laplace transform for causal functions; i.e., $x(t) = 0$ for $t < 0$.

Property	$x(t)$	$\mathbf{X}(s) = \mathcal{L}[x(t)]$
1. Multiplication by constant	$K x(t)$	$K \mathbf{X}(s)$
2. Linearity	$K_1 x_1(t) + K_2 x_2(t)$	$K_1 \mathbf{X}_1(s) + K_2 \mathbf{X}_2(s)$
3. Time scaling	$x(at), a > 0$	$\frac{1}{a} \mathbf{X}\left(\frac{s}{a}\right)$
4. Time shift	$x(t-T) u(t-T)$	$e^{-Ts} \mathbf{X}(s), T \geq 0$
5. Frequency shift	$e^{-at} x(t)$	$\mathbf{X}(s+a)$
6. Time 1st derivative	$x' = \frac{dx}{dt}$	$s \mathbf{X}(s) - x(0^-)$
7. Time 2nd derivative	$x'' = \frac{d^2x}{dt^2}$	$s^2 \mathbf{X}(s) - sx(0^-) - x'(0^-)$
8. Time integral	$\int_{0^-}^t x(t') dt'$	$\frac{1}{s} \mathbf{X}(s)$
9. Frequency derivative	$t x(t)$	$-\frac{d}{ds} \mathbf{X}(s) = -\mathbf{X}'(s)$
10. Frequency integral	$\frac{x(t)}{t}$	$\int_s^\infty \mathbf{X}(s') ds'$
11. Initial value	$x(0^+) = \lim_{t \rightarrow 0^+} x(t)$	$\lim_{s \rightarrow \infty} s \mathbf{X}(s)$
12. Final value	$\lim_{t \rightarrow \infty} x(t) = x(\infty)$	$\lim_{s \rightarrow 0} s \mathbf{X}(s)$
13. Convolution	$x_1(t) * x_2(t)$	$\mathbf{X}_1(s) \mathbf{X}_2(s)$

Notes:

- The initial value theorem only works if $\mathbf{X}(s)$ is strictly proper since the limit $\lim_{s \rightarrow \infty} s \mathbf{X}(s)$ does not exist otherwise.
- The final value theorem only works when $s \mathbf{X}(s)$ does not have any poles in the right-half plane or on the imaginary axis. In other words, any poles of $s \mathbf{X}(s)$ must be in the open left-half plane.

$x_2(t) = A x_1[a(t-t_0)]$

A = amplitude scaling

t_0 = time shift

a = time scaling / reversal

- $x_1(t-t_0) \rightarrow$ shift right by t_0
- $x_1(t+t_0) \rightarrow$ shift left by t_0
- $x_1(at)$
 - $|a| > 1 \rightarrow$ compress
 - $0 < |a| < 1 \rightarrow$ expand
- $x_1(-t) \rightarrow$ reflect across $t=0$
- $A x_1(t) \rightarrow$ amplitude scale by A
 - $A > 1 \rightarrow$ taller
 - $0 < A < 1 \rightarrow$ shorter
 - $A < 0 \rightarrow$ flip vertically and scale

- Rewrite the inside as $a(t-t_0)$.
- Shift by t_0 first.
- Scale time by $1/a$.
- If $a < 0$, reflect in time.
- Apply outside amplitude A last.

Shift -> Scale -> Reflect -> Amplitude

Euler formulas

$$e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

$$e^{-j\omega t} = \cos(\omega t) - j \sin(\omega t)$$

Integration by Parts

$$\int u dv = uv - \int v du$$

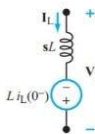
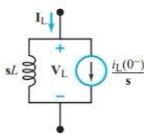
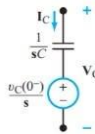
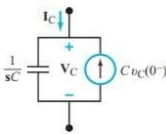
$$V_1 = I_1 A - I_2 B$$

$$V_2 = -I_1 C + I_2 D$$

$$\Delta = \begin{vmatrix} A & -B \\ -C & D \end{vmatrix} = AD - BC$$

$$\Delta I_2 = \begin{vmatrix} A & V_1 \\ -C & V_2 \end{vmatrix} = AV_2 + CV_1$$

$$I_2 = \frac{\Delta I_2}{\Delta}$$

Time-Domain	s-Domain
Resistor  $v = Ri$	 $V = RI$
Inductor  $v_L = L \frac{di_L}{dt}$ $i_L = \frac{1}{L} \int_{0^-}^t v_L dt' + i_L(0^-)$	 $V_L = sLI_L - L i_L(0^-)$ OR  $I_L = \frac{V_L}{sL} + \frac{i_L(0^-)}{s}$
Capacitor  $i_C = C \frac{dv_C}{dt}$ $v_C = \frac{1}{C} \int_{0^-}^t i_C dt + v_C(0^-)$	 $V_C = \frac{I_C}{sC} + \frac{v_C(0^-)}{s}$ OR  $I_C = sC V_C - C v_C(0^-)$

Cosine/Sine

- Fourier Series (Trigonometric Form)
$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)$$
- Angular Frequency
$$\omega_0 = \frac{2\pi}{T_0}$$
- Fourier Coefficients
$$a_0 = \frac{1}{T_0} \int_{T_0} x(t) dt$$

$$a_n = \frac{2}{T_0} \int_{T_0} x(t) \cos(n\omega_0 t) dt$$

$$b_n = \frac{2}{T_0} \int_{T_0} x(t) \sin(n\omega_0 t) dt$$
- Types of Symmetry
 - Odd: $x(t) = -x(-t)$ (upside-down mirror through origin)
$$a_0 = 0, a_n = 0$$
Odd $\rightarrow$ only sin
 - Even: $x(t) = x(-t)$ (mirror across vertical axis)
$$b_n = 0$$
Even $\rightarrow$ only cos
 - Zero Average (No DC): $a_0 = 0$
positive and negative parts are equal over one period

All integrals are over one period, T_0 .

Odd $\rightarrow$ sine only | Even $\rightarrow$ cosine only | If average = 0, then $a_0 = 0$

Amplitude/Phase

$$x(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \phi_n)$$

$$c_n e^{j\phi_n} = a_n - jb_n$$

$$c_n = \sqrt{a_n^2 + b_n^2}$$

$$\phi_n = \begin{cases} -\tan^{-1}(b_n/a_n), & a_n > 0 \\ \pi - \tan^{-1}(b_n/a_n), & a_n < 0 \end{cases}$$

$$a_n = c_n \cos \phi_n; b_n = -c_n \sin \phi_n$$

Exponential

$$x(t) = \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t}$$

$$\mathbf{x}_n = |\mathbf{x}_n| e^{j\phi_n}; \mathbf{x}_{-n} = \mathbf{x}_n^*; \phi_{-n} = -\phi_n$$

$$|\mathbf{x}_n| = c_n/2; \mathbf{x}_0 = c_0$$

$$\mathbf{x}_n = \frac{1}{T_0} \int_{T_0} x(t) e^{-jn\omega_0 t} dt$$

$$\mathbf{x}_n = \frac{1}{2}(a_n - jb_n)$$

Fourier transform properties.

Property	$x(t)$	$\mathbf{X}(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$
1. Multiplication by constant	$K x(t)$	$K \mathbf{X}(\omega)$
2. Linearity	$K_1 x_1(t) + K_2 x_2(t)$	$K_1 \mathbf{X}_1(\omega) + K_2 \mathbf{X}_2(\omega)$
3. Time scaling	$x(at)$	$\frac{1}{ a } \mathbf{X}\left(\frac{\omega}{a}\right)$
4. Time shift	$x(t-t_0)$	$e^{-j\omega t_0} \mathbf{X}(\omega)$
5. Frequency shift	$e^{j\omega_0 t} x(t)$	$\mathbf{X}(\omega - \omega_0)$
6. Time 1st derivative	$x' = \frac{dx}{dt}$	$j\omega \mathbf{X}(\omega)$
7. Time n-th derivative	$\frac{d^n x}{dt^n}$	$(j\omega)^n \mathbf{X}(\omega)$
8. Time integral	$\int_{-\infty}^t x(\tau) d\tau$	$\frac{\mathbf{X}(\omega)}{j\omega} + \pi \mathbf{X}(0) \delta(\omega)$, where $\mathbf{X}(0) = \int_{-\infty}^{\infty} x(t) dt$
9. Frequency derivative	$t^n x(t)$	$(j)^n \frac{d^n \mathbf{X}(\omega)}{d\omega^n}$
10. Modulation	$x(t) \cos \omega_0 t$	$\frac{1}{2} [\mathbf{X}(\omega - \omega_0) + \mathbf{X}(\omega + \omega_0)]$
11. Convolution in t	$x_1(t) * x_2(t)$	$\mathbf{X}_1(\omega) \mathbf{X}_2(\omega)$
12. Convolution in ω	$x_1(t) x_2(t)$	$\frac{1}{2\pi} \mathbf{X}_1(\omega) * \mathbf{X}_2(\omega)$
13. Conjugate symmetry		$\mathbf{X}(-\omega) = \mathbf{X}^*(\omega)$

Common Convolutions

- $u(t) * u(t) = t u(t)$
- $e^{at} u(t) * u(t) = \left(\frac{e^{at} - 1}{a} \right) u(t)$
- $e^{at} u(t) * e^{bt} u(t) = \left(\frac{e^{at} - e^{bt}}{a - b} \right) u(t), \quad a \neq b$
- $e^{at} u(t) * e^{at} u(t) = t e^{at} u(t)$
- $t e^{at} u(t) * e^{bt} u(t) = \frac{e^{bt} - e^{at}}{(a - b)^2} + \frac{(a - b)t e^{at}}{(a - b)^2} u(t), \quad a \neq b$
- $t e^{at} u(t) * e^{at} u(t) = \frac{1}{2} t^2 e^{at} u(t)$
- $\delta(t - T_1) * \delta(t - T_2) = \delta(t - T_1 - T_2)$

Convolution with a Ramp

$$r(t) = t u(t)$$

$$u(t) * r(t) = u(t) * t u(t) = \frac{1}{2} t^2 u(t)$$

$$u(t) * r(t) = \frac{1}{2} r_2(t)$$

$$r_2(t) = t^2 u(t)$$

Convolving with $u(t)$ = integrate from 0 to t

$$u(t) * r(t) = \int_0^t \tau d\tau = \frac{t^2}{2} u(t)$$

Total, Natural, and Forced Response

Total response = Natural response + Forced response

$$y_{\text{total}}(t) = y_n(t) + y_f(t)$$

$$Y(s) = Y_n(s) + Y_f(s)$$

①	Natural response (zero-input response) - Caused by initial conditions / stored energy - Set input to zero: $x(t) = 0$ - Comes from capacitor and inductor initial conditions
②	Forced response (zero-state response) - Caused by the input - Assume zero initial conditions - Response due only to $x(t)$
③	Total response - Actual output of the system - Add natural + forced

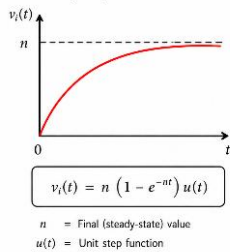
Natural = stored energy Forced = input

$$\begin{aligned} ax + by &= e \\ cx + dy &= f \end{aligned} \quad \Delta = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \neq 0 \quad \text{IM}$$

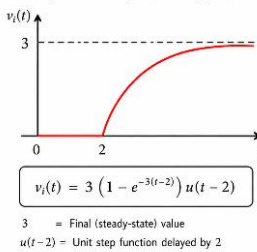
$$\Delta_x = \begin{vmatrix} e & b \\ f & d \end{vmatrix} = ed - bf, \quad \Delta_y = \begin{vmatrix} a & e \\ c & f \end{vmatrix} = af - ec$$

$$x = \frac{\Delta_x}{\Delta}, \quad y = \frac{\Delta_y}{\Delta}$$

Step Response (General)



Specific Example ($n = 3, t_0 = 2$)



$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

- $u(t)$ turns the response on at $t = 0$.
- $u(t-2)$ delays the response until $t = 2$.

BIBO Stability from Pole Locations

Continuous-time LTI system is BIBO stable if all poles satisfy $\text{Re}\{s\} < 0$

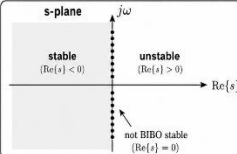
Stable	Unstable	Imaginary-Axis Poles
All poles in LEFT half-plane $\text{Re}\{s\} < 0$ Example: $s = -1 \pm j$ Time response decays: $e^{-t} \cos(t), e^{-t} \sin(t)$ Result: BIBO stable	Any pole in RIGHT half-plane $\text{Re}\{s\} > 0$ Example: $s = 1 + j$ Time response grows: $e^t e^{jt}$ Result: NOT BIBO stable	Poles on $j\omega$ -axis $\text{Re}\{s\} = 0$ Example: $s = \pm j$ Oscillatory, does not decay Result: NOT BIBO stable

Quick check for poles $s = \sigma + j\omega$:

$$\sigma < 0 \rightarrow \text{stable}$$

$$\sigma = 0 \rightarrow \text{not BIBO stable}$$

$$\sigma > 0 \rightarrow \text{unstable}$$



Odd/Even Symmetry

$$x(t) = x_e(t) + x_o(t)$$

$$x_e(t) = \frac{x(t) + x(-t)}{2}, \quad x_o(t) = \frac{x(t) - x(-t)}{2}$$

Laplace Transform:

$$X(s) = L[x(t)] = \int_0^\infty x(t) e^{-st} dt$$

$$s = \sigma + j\omega; \quad |s| = \sqrt{\sigma^2 + \omega^2}$$

Partitions:

$$V(s) = \frac{RC}{1+sRC} V(0^-) + \frac{1}{1+sRC} V_i(s)$$

ZIR ZSR

$$\text{ZIR: set } V_i(s) = 0$$

ZSR: Zero initial conditions

Transient: Decay to 0 overtime

Steady-State: Remains overtime

Natural Response: Terms that depend only on MODES

Forced Response: All remaining terms

Fourier Methods Through LTI Circuits

1) Fundamental Frequency	2) General Fourier-Series Input
$\omega_0 = \frac{2\pi}{T_0}$ $\omega_n = n\omega_0 = \frac{2n\pi}{T_0}$	$x(t) = a_0 + \sum_{n=1}^\infty A_n \cos(n\omega_0 t + \phi_n)$
3) Output Through an LTI Circuit	
$y(t) = a_0 H(0) + \sum_{n=1}^\infty A_n H(n\omega_0) \cos(n\omega_0 t + \phi_n + \angle H(n\omega_0))$	
4) Harmonic Update Rules	
$A_n \rightarrow A_n H(n\omega_0) $	$\phi_n \rightarrow \phi_n + \angle H(n\omega_0)$
5) RL Low-Pass	6) RC Low-Pass
$H(\omega) = \frac{R}{R + j\omega L}$ $ H(\omega) = \frac{R}{\sqrt{R^2 + (\omega L)^2}}$ $\angle H(\omega) = -\tan^{-1}\left(\frac{\omega L}{R}\right)$ $ H(n\omega_0) = \frac{R}{\sqrt{R^2 + (n\omega_0 L)^2}}$ $\angle H(n\omega_0) = -\tan^{-1}\left(\frac{n\omega_0 L}{R}\right)$	$H(\omega) = \frac{1}{1 + j\omega RC}$ $ H(\omega) = \frac{1}{\sqrt{1 + (\omega RC)^2}}$ $\angle H(\omega) = -\tan^{-1}(\omega RC)$ $ H(n\omega_0) = \frac{1}{\sqrt{1 + (n\omega_0 RC)^2}}$ $\angle H(n\omega_0) = -\tan^{-1}(n\omega_0 RC)$
7) RC High-Pass	8) Final Template
$H(\omega) = \frac{j\omega RC}{1 + j\omega RC}$ $ H(\omega) = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}}$ $\angle H(\omega) = 90^\circ - \tan^{-1}(\omega RC)$ $ H(n\omega_0) = \frac{n\omega_0 RC}{\sqrt{1 + (n\omega_0 RC)^2}}$ $\angle H(n\omega_0) = 90^\circ - \tan^{-1}(n\omega_0 RC)$ $\angle H(\omega) = \frac{\pi}{2} - \tan^{-1}(\omega_0 RC)$	If $x(t) = \sum_{n=1}^\infty A_n \cos(n\omega_0 t + \phi_n)$, then $y(t) = \sum_{n=1}^\infty A_n H(n\omega_0) \cos(n\omega_0 t + \phi_n + \angle H(n\omega_0))$

Energy and Power

$$E[x(t)] = \int_{-\infty}^\infty |x(t)|^2 dt$$

$$P_{av} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

BIBO

Casual System: Present value of output $y(t)$

Can only depend on present and past values

Ex: $u(t), 2x(t-1) + x(t)$

Not Casual: $3x(t+2)$ it must know input at -2

BIBO stable if

$$\int_{-\infty}^\infty |h(t)| dt = \text{finite} < \infty$$

Frequency Response: $\hat{H}(\omega)$

$$y(t) = \hat{H}(\omega) e^{j\omega t}; \quad x(t) = e^{j\omega t}$$

$$\frac{d}{dt} \rightarrow j\omega; \quad \frac{d^2}{dt^2} \rightarrow (j\omega)^2$$

$$h(t) = e^{-t} u(t); \quad \hat{H}(\omega) = \int_{-\infty}^\infty h(t) e^{-j\omega t} dt = \frac{1}{1 + j\omega}$$

3-11.2 Natural / Forced Partition

If we partition the total response into a

• **Natural response** = all terms that vary with t according to the system's modes (in the present case, the system has only one mode, namely e^{-3t}), and a

• **Forced response** = all remaining terms,

• **Zero-input response (ZIR)** = response of the system in the absence of an input excitation ($V_i(s) = 0$).

• **Zero-state response (ZSR)** = response of the system when it is in zero state (zero initial conditions, $A_c = v(0^-) = 0$).

Average Power of a Sinusoid

$$x_1(t) = A \cos(n\omega_0 t + \phi), \quad \omega_0 \neq 0, \quad n > 0$$

$$P_{x1} = \frac{A^2}{2}$$

Power of a Complex Exponential

$$x_2(t) = A e^{j(\omega_0 t + \phi)}$$

$$P_{x2} = A^2$$

The Fourier Transform

$$\hat{X}(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^\infty x(t) e^{-j\omega t} dt$$

$$x(t) = \mathcal{F}^{-1}[\hat{X}(\omega)] = \frac{1}{2\pi} \int_{-\infty}^\infty \hat{X}(\omega) e^{j\omega t} d\omega$$

Natural response: set input to 0, keep initial conditions

Forced response: keep input, set output initial conditions to 0

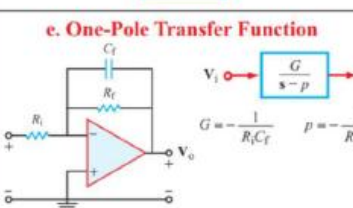
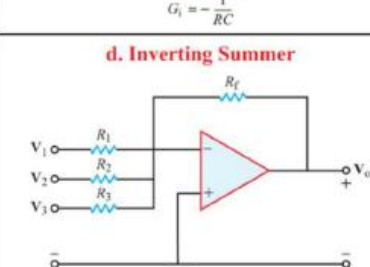
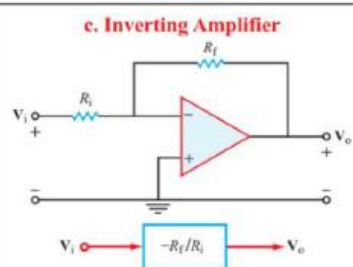
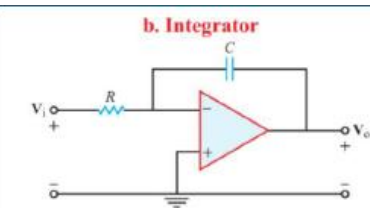
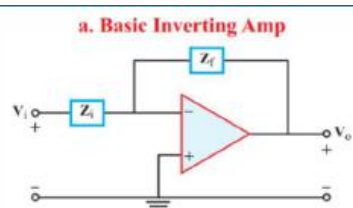
3 Versions of Power Formula

$$P_x = \frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt = a_0^2 + \sum_{n=1}^\infty (a_n^2 + b_n^2)/2, \quad (5.58a)$$

$$P_x = \frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt = c_0^2 + \sum_{n=1}^\infty c_n^2/2, \quad (5.58b)$$

$$P_x = \frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt = \sum_{n=-\infty}^\infty |x_n|^2. \quad (5.58c)$$

$$P_{x+y} = \frac{c_n^2}{2} + \frac{c_m^2}{2}$$



Parseval's Theorem:

$$\text{Instantaneous Power} = p(t) = |x(t)|^2 = x(t)x^*(t)$$

Total Energy E Expended from t_1 to t_2 :

$$E = \int_{t_1}^{t_2} p(t) dt = \int_{t_1}^{t_2} |x(t)|^2 dt = \int_{t_1}^{t_2} x(t)x^*(t) dt$$

$$\text{Average Power: } P_{av} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} p(t) dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} |x(t)|^2 dt$$

Transient / steady-state partition

In many practical situations, the system analyst or designer may be particularly interested in partitioning the response into a

- **Transient response** = all terms that decay to zero as $t \rightarrow \infty$, and a
- **Steady-state response** = all terms that remain after the demise of the transient response.

Partitioning the expression for $v(t)$ along those lines gives

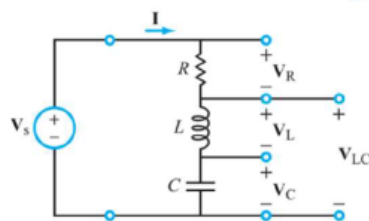
$$v(t) = \underbrace{[14e^{-3t} - 18e^{-2t}] u(t)}_{\text{Transient response}} + \underbrace{6u(t)}_{\text{Steady-state response}}$$

The terms included in the first square bracket decay exponentially to zero as $t \rightarrow \infty$, leaving behind the steady-state component $6u(t)$. The steady-state component need not be a constant (dc value). In fact, if the input function is a sinusoid, the steady-state response will also be a sinusoid at the same frequency.

- The power associated with the zero frequency terms (a_0 , c_0 , or $|x_0|$) is called the **dc power**
- The remaining power associated with the harmonics is referred to as the **ac power**
- The **ac power fraction** is the ratio of the ac power and the sum of the ac power and dc power

bandstop filtering, we choose V_L

Bandpass Filter



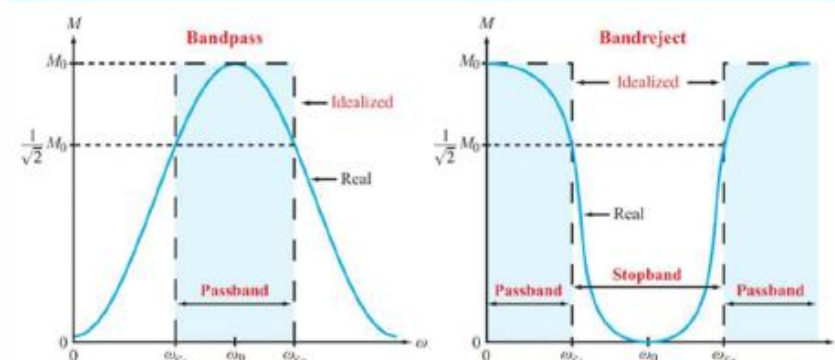
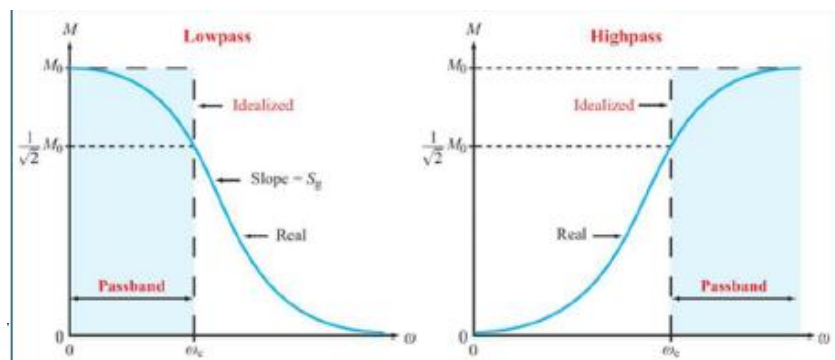
$$B = \omega_{c2} - \omega_{c1} = \frac{R}{L} \quad \text{BANDWIDTH}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Maximum value at $\omega_0 = 1$

$$\text{Half power} \Rightarrow M_{BP}(\omega) = \frac{1}{\sqrt{2}}$$

$x(t)$	$\longleftrightarrow$	$X(\omega) = \mathcal{F}[x(t)]$	$ X(\omega) $
1.	$\delta(t) \longleftrightarrow$	1	
1a.	$\delta(t - t_0) \longleftrightarrow$	$e^{-j\omega t_0}$	
2.	$1 \longleftrightarrow$	$2\pi \delta(\omega)$	
3.	$u(t) \longleftrightarrow$	$\pi \delta(\omega) + 1/j\omega$	
4.	$\text{sgn}(t) \longleftrightarrow$	$2/j\omega$	
5.	$\text{rect}(t/\tau) \longleftrightarrow$	$\tau \text{sinc}(\omega\tau/2)$	
6.	$\frac{e^{-t^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}} \longleftrightarrow$	$e^{-\omega^2\sigma^2/2}$	
7a.	$e^{-at}u(t) \longleftrightarrow$	$1/(a + j\omega)$	
7b.	$e^{at}u(-t) \longleftrightarrow$	$1/(a - j\omega)$	
8.	$\cos \omega_0 t \longleftrightarrow$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	
9.	$\sin \omega_0 t \longleftrightarrow$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$	
10. $e^{j\omega_0 t}$	$\longleftrightarrow$	$2\pi \delta(\omega - \omega_0)$	
11. $t e^{-at} u(t)$	$\longleftrightarrow$	$1/(a + j\omega)^2$	
12a. $[e^{-at} \sin \omega_0 t] u(t)$	$\longleftrightarrow$	$\omega_0 / [(a + j\omega)^2 + \omega_0^2]$	
12b. $[\sin \omega_0 t] u(t)$	$\longleftrightarrow$	$(\pi/2j)[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] + [\omega_0^2/(\omega_0^2 - \omega^2)]$	
13a. $[e^{-at} \cos \omega_0 t] u(t)$	$\longleftrightarrow$	$(a + j\omega)/[(a + j\omega)^2 + \omega_0^2]$	
13b. $[\cos \omega_0 t] u(t)$	$\longleftrightarrow$	$(\pi/2)[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] + [j\omega/(\omega_0^2 - \omega^2)]$	



Fourier transform properties.

Property	$x(t) \longleftrightarrow$	$X(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$
1. Multiplication by constant	$K x(t) \longleftrightarrow$	$K X(\omega)$
2. Linearity	$K_1 x_1(t) + K_2 x_2(t) \longleftrightarrow$	$K_1 X_1(\omega) + K_2 X_2(\omega)$
3. Time scaling	$x(at) \longleftrightarrow$	$\frac{1}{ a } X\left(\frac{\omega}{a}\right)$
4. Time shift	$x(t - t_0) \longleftrightarrow$	$e^{-j\omega t_0} X(\omega)$
5. Frequency shift	$e^{j\omega_0 t} x(t) \longleftrightarrow$	$X(\omega - \omega_0)$
6. Time 1st derivative	$x' = \frac{dx}{dt} \longleftrightarrow$	$j\omega X(\omega)$
7. Time nth derivative	$\frac{d^n x}{dt^n} \longleftrightarrow$	$(j\omega)^n X(\omega)$
8. Time integral	$\int_{-\infty}^t x(\tau) d\tau \longleftrightarrow$	$\frac{X(\omega)}{j\omega} + \pi X(0) \delta(\omega)$, where $X(0) = \int_{-\infty}^{\infty} x(t) dt$
9. Frequency derivative	$t^n x(t) \longleftrightarrow$	$(j)^n \frac{d^n X(\omega)}{d\omega^n}$
10. Modulation	$x(t) \cos \omega_0 t \longleftrightarrow$	$\frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$
11. Convolution in t	$x_1(t) * x_2(t) \longleftrightarrow$	$X_1(\omega) X_2(\omega)$
12. Convolution in omega	$x_1(t) x_2(t) \longleftrightarrow$	$\frac{1}{2\pi} X_1(\omega) * X_2(\omega)$
13. Conjugate symmetry		$X(-\omega) = X^*(\omega)$

Bandpass Filter

$$\hat{H}_R(\omega) = \frac{V_R}{V_s} = \frac{IR}{V_s} = \frac{j\omega RC}{j\omega RC + (1 - \omega^2 LC)}$$
 when $\omega \rightarrow 0 \Rightarrow |\hat{H}_R(\omega)| \rightarrow 0$
when $\omega \rightarrow \infty \Rightarrow |\hat{H}_R(\omega)| \rightarrow 0$

HIGH-PASS

$$\hat{H}_L(\omega) = \frac{V_L}{V_s} = \frac{(j\omega L)I}{V_s} = \frac{(j\omega L)(j\omega C)}{j\omega RC + (1 - \omega^2 LC)}$$
 when $\omega \rightarrow 0 \Rightarrow |\hat{H}_L(\omega)| \rightarrow 0$
when $\omega \rightarrow \infty \Rightarrow |\hat{H}_L(\omega)| \rightarrow 1$

LOW-PASS FILTER

$$\hat{H}_C(\omega) = \frac{V_C}{V_s} = \frac{I}{j\omega C} \times \frac{1}{V_s} = \frac{1}{j\omega RC + (1 - \omega^2 LC)}$$
 when $\omega \rightarrow 0 \Rightarrow |\hat{H}_C(\omega)| \rightarrow 1$
when $\omega \rightarrow \infty \Rightarrow |\hat{H}_C(\omega)| \rightarrow 0$

S.No.	Form of the rational function	Form of the partial fraction
1.	$\frac{px+q}{(x-a)(x-b)}, a \neq b$	$\frac{A}{x-a} + \frac{B}{x-b}$
2.	$\frac{px+q}{(x-a)^2}$	$\frac{A}{x-a} + \frac{B}{(x-a)^2}$
3.	$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)}$	$\frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$
4.	$\frac{px^2+qx+r}{(x-a)^2(x-b)}$	$\frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$
5.	$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)}$	$\frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$

where $x^2 + bx + c$ cannot be factorised further

$$M = |\hat{H}(\omega)| = \frac{(\omega / \omega_c)}{\sqrt{1 + (\omega / \omega_c)^2}}$$

$$M[\text{dB}] = 20 \log M = 20 \log(\omega / \omega_c) - 20 \log \sqrt{1 + (\omega / \omega_c)^2}$$

1 2

$$\hat{H}(\omega) = \frac{j(\omega / \omega_c)}{1 + j(\omega / \omega_c)}$$

$$\phi(\omega) = \phi_{\text{numerator}} - \phi_{\text{denominator}} = 90^\circ - \tan^{-1}(\omega / \omega_c)$$

1 2

Step 1: Express $v_s(t)$ in terms of an amplitude/phase Fourier Series:

$$v_s(t) = a_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \phi_n)$$
 (1)

Step 2: Determine the transfer function of the circuit at frequency ω :

$$\hat{H}(\omega) = V_{out} \text{ when } v_s = \cos \omega t$$

Step 3: Adjust each term in equation (1) by the transfer function from Step 2:

$$v_{out}(t) = a_0 \hat{H}(0) + \sum_{n=1}^{\infty} c_n |\hat{H}(n\omega_0)| \cos(n\omega_0 t + \phi_n + \angle \hat{H}(n\omega_0))$$

Factor	Bode Magnitude	Bode Phase
Constant K	$20 \log K$ 0 dB	$\pm 180^\circ$ if $K < 0$ 0° if $K > 0$
Zero @ Origin $(j\omega)^N$	slope = $20N$ dB/decade	$(90N)^\circ$
Pole @ Origin $(j\omega)^{-N}$	slope = $-20N$ dB/decade	$(-90N)^\circ$
Simple Zero $(1 + j\omega/\omega_c)^N$	slope = $20N$ dB/decade	$(90N)^\circ$
Simple Pole $(1 + j\omega/\omega_c)^{-N}$	slope = $-20N$ dB/decade	$(-90N)^\circ$
Quadratic Zero $[1 + j2\zeta\omega/\omega_c + (j\omega/\omega_c)^2]^N$	slope = $40N$ dB/decade	$(180N)^\circ$
Quadratic Pole $[1 + j2\zeta\omega/\omega_c + (j\omega/\omega_c)^2]^{-N}$	slope = $-40N$ dB/decade	$(-180N)^\circ$

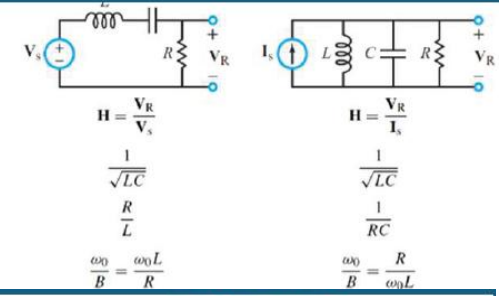
RLC circuit

Transfer function

Resonant frequency, ω_0

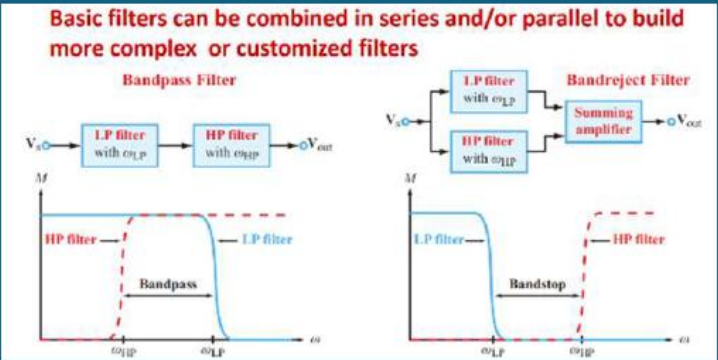
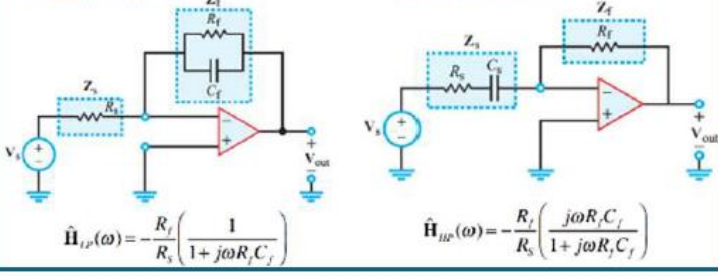
Bandwidth, B

Quality factor, Q



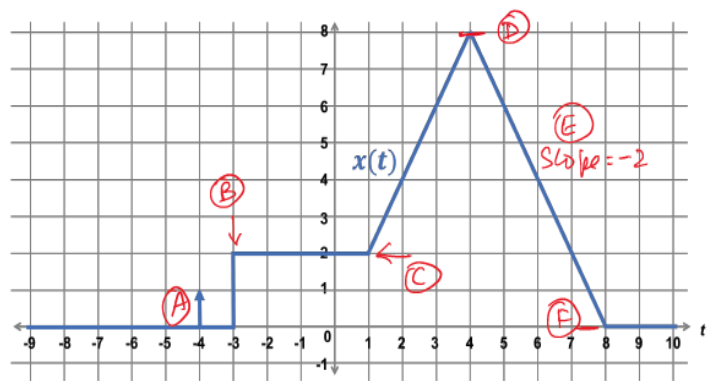
ACTIVE LOW-PASS FILTER

ACTIVE HIGH-PASS FILTER



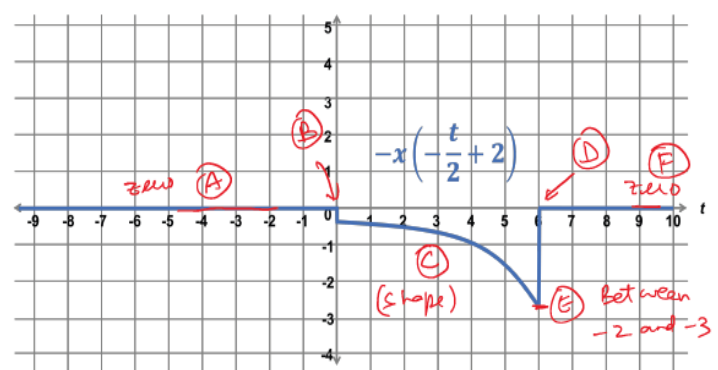
1. (15 points) Waveforms & Signal Transformations

a. (7 points) Plot the signal $x(t) = 2u(t+3) + \delta(t+4) + 2r(t-1) - 4r(t-4) + 2r(t-8)$. Your final answer should be plotted below and will be the only response that will be graded.



1 point for trying. 1 point for A through F.

b. (8 points) If $x(t) = e^{-t}[u(t+1) - u(t-2)]$, plot $y(t) = -x(-\frac{t}{2} + 2)$ on the graph below. Your final answer should be plotted below and will be the only response that will be graded.



3. (15 points) Signal Properties

a. (4 points) A signal $x(t)$ has energy $E[x(t)]$. What is the energy of the signal $cx(at-b)$?

$$E[x(t)] = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$E[cx(at-b)] = \int_{-\infty}^{\infty} |c x(at-b)|^2 dt = |c|^2 \int_{-\infty}^{\infty} |x(at-b)|^2 dt = |c|^2 \int_{-\infty}^{\infty} |x(u)|^2 \frac{du}{a} = \frac{|c|^2}{|a|} \int_{-\infty}^{\infty} |x(u)|^2 du = \frac{|c|^2}{|a|} E[x(t)]$$

Let $u = at-b$
 $\Rightarrow du = a dt$

Grading: .1 pt for trying
 +1 pt for (A)
 +4 pts correct answer

b. (4 points) Calculate the integral $I = \int_{-\infty}^{\infty} t^5 \delta(3t-5) dt$.

$$I = \int_{-\infty}^{\infty} t^5 \delta(3t-5) dt = \int_{-\infty}^{\infty} t^5 \frac{1}{3} \delta(t-\frac{5}{3}) dt$$

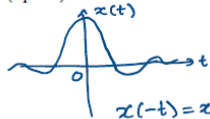
(A): Time scaling of $\delta(\cdot)$

$$= \frac{1}{3} \int_{-\infty}^{\infty} t^5 \delta(t-\frac{5}{3}) dt = \frac{1}{3} (\frac{5}{3})^5$$

(B): Sampling property.

Grading: .1 pt for trying
 +1 pt for (A) & (B) each
 +4 pts correct answer

c. (3 points) Determine whether the signal $x(t) = \frac{\sin(3t)}{3t}$ has even, odd, or no symmetry.



$x(-t) = x(t) \Rightarrow$ Even symmetry

Grading: .1 pt for trying
 +3 pts correct answer

d. (4 points) Determine the amplitude, phase, and time period of the signal $x(t) = (3+j4)e^{j3t+4}$.

$$x(t) = \sqrt{3^2+4^2} e^{j \tan^{-1}(\frac{4}{3})} e^{j3t} e^4$$

Amplitude = $\sqrt{3^2+4^2} \times e^4 = 5e^4$ (A)

Phase = $\tan^{-1}(\frac{4}{3})$ (B)

Frequency $\omega = 3 = \frac{2\pi}{T_0}$
 $\Rightarrow$ Period $T_0 = \frac{2\pi}{3}$ (C)

Grading: 1 pt for trying + 1 pt each correct (A), (B), (C).

3. (20 points) System Properties

a. (5 points) Does the system $y(t) = e^{-j2\pi t} \sin(t)$ have the causality property? Explain your reasoning.

Notice that the output of the system $y(t)$ at time t depends only on the current value of the input $x(t)$. It does not depend on any future values of the input. Therefore, the system is causal.

(A)

(B)

1 point for trying. 2 points for A, B.

b. (5 points) Is the system $y(t) = \int_{t-1}^{t+3} x(\tau) d\tau$ time invariant? Explain your reasoning.

If the input is $x_1(t)$, the output is $y_1(t) = \int_{t-1}^{t+3} x_1(\tau) d\tau$.

Delaying this output by T seconds gives $y_1(t-T) = \int_{t-T-1}^{t-T+3} x_1(\tau) d\tau$.

If the input is $x_2(t) = x_1(t-T)$, then the output is

$$y_2(t) = \int_{t-1}^{t+3} x_2(\tau) d\tau$$

Substituting $\tau - T = u$, the limits go from $t-1-T$ to $t+3-T$.

$$y_2(t) = \int_{t-T-1}^{t-T+3} x_1(u) du = y_1(t-T)$$

Therefore, the system is time invariant.

(B)

1 point for trying. 1 point for A-D.

c. (5 points) Is the system $y(t) = 10 e^{j\theta x(t)^2}$ BIBO stable? Explain your reasoning.

The system is BIBO stable if the output $|y(t)| = |10 e^{j\theta x(t)^2}|$ is bounded when the input $|x(t)|$ is bounded. If $|x(t)| < C$ is bounded, then $|x(t)|^2 < C^2$ is also bounded. Note that $|e^{j\theta}| = 1$ for all finite values of θ . Consequently, $|y(t)| = |10 e^{j\theta x(t)^2}| = 10 |e^{j\theta x(t)^2}| = 10$ is bounded by 10 if $x(t)$ is bounded. Therefore, the system is BIBO stable.

(B)

d. (5 points) Is the system defined by $y(t) = \frac{\sin(100t)}{2t} x(t)$ linear? Explain your answer.

Let the outputs for inputs $x_1(t)$ and $x_2(t)$ be

$$y_1(t) = \frac{\sin(100t)}{2t} x_1(t) \text{ and}$$

$$y_2(t) = \frac{\sin(100t)}{2t} x_2(t)$$

The output for $x(t) = a x_1(t) + b x_2(t)$ is

(A)

$$y(t) = \frac{\sin(100t)}{2t} x(t)$$

$$y(t) = \frac{\sin(100t)}{2t} [a x_1(t) + b x_2(t)]$$

$$= a \frac{\sin(100t)}{2t} x_1(t) + b \frac{\sin(100t)}{2t} x_2(t)$$

$$= a y_1(t) + b y_2(t)$$

Therefore, the system is linear.

(D)

(C)

1 point for trying. 1 point for A-D.

4. (20 points) Convolution Integral

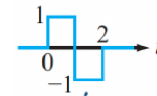
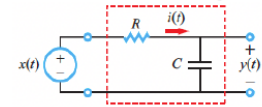
a. (5 points) Compute the following convolution without computing any integrals $y(t) = u(t+1) * [\delta(t-2) + u(t-3) + r(t-1)]$, where $r(t)$ is the ramp function.

$$y(t) = u(t+1) * \delta(t-2) + u(t+1) * u(t-3) + u(t+1) * r(t-1)$$

$$= u(t-1) + r(t-2) + \frac{1}{2} t^2 u(t)$$

Grading: 2 pts for trying
 +1 pt each correct (A), (B), (C).

b. (7 points) Write down the step response $y_{step}(t)$ of the following circuit with $RC = 1$, and zero initial conditions. Then calculate the response to the input waveform indicated below in terms of $y_{step}(t)$.



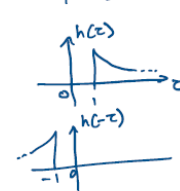
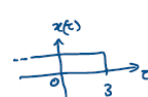
$$x(t) = u(t) - 2u(t-1) + u(t-2)$$

$$\Rightarrow y(t) = y_{step}(t) - 2y_{step}(t-1) + y_{step}(t-2)$$

where $y_{step}(t) = (1 - e^{-t}) u(t)$

Grading: 2 pts for trying
 +1 pt each correct (A), (B), (C), (D), (E).

c. (8 points) Using explicit integration and showing all steps, calculate the output $y(t)$ of an LTI system with impulse response $h(t) = e^{-2t} u(t-1)$ for the input $x(t) = u(3-t)$.



$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

Choose

$$= \int_{-\infty}^{\infty} u(3-\tau) e^{-2(t-\tau)} u(t-\tau-1) d\tau$$

$3-\tau > 0 \Rightarrow \tau < 3$
 $t-\tau-1 > 0 \Rightarrow \tau < t-1$

when $t < 4 \Rightarrow$ limits are $-\infty$ to $t-1$ I_1
 when $t > 4 \Rightarrow$ limits are $-\infty$ to 3 I_2

$$I_1 = u(4-t) \int_{-\infty}^{t-1} e^{-2(t-\tau)} d\tau, I_2 = u(t-4) \int_{-\infty}^3 e^{-2(t-\tau)} d\tau$$

$$= u(4-t) e^{-2t} \int_{-\infty}^{t-1} e^{2\tau} d\tau + u(t-4) e^{-2t} \int_{-\infty}^3 e^{2\tau} d\tau$$

$$= u(4-t) e^{-2t} \left\{ \frac{e^{2(t-1)}}{2} - 0 \right\} + u(t-4) e^{-2t} \left\{ \frac{e^6}{2} - 0 \right\}$$

$$= u(4-t) \frac{e^{-2}}{2} + u(t-4) \frac{e^{-2+6}}{2}$$

Grading: 2 pts for trying
 +1 pt each correct (A)-(F)

5. (15 points) System Response

Consider the LTI system given by the differential equation

$$3 \frac{dy(t)}{dt} + 4y(t) = -\frac{dx(t)}{dt}$$

Assume that $y(0^-) = 0$.

a. (7 points) Find the step response $y_{step}(t)$ of the above system.

To find the step response, give the system an input $x(t) = u(t)$.

$$\frac{dx(t)}{dt} = \delta(t)$$

The differential equation reduces to

$$3 \frac{dy(t)}{dt} + 4y(t) = -\delta(t)$$

$$\frac{dy(t)}{dt} + \frac{4}{3}y(t) = -\frac{1}{3}\delta(t)$$

This is a first-order linear differential equation. The integrating factor is $e^{\int \frac{4}{3} dt} = e^{\frac{4}{3}t}$. Multiplying by the integrating factor and simplifying.

$$\frac{dy(t)}{dt} e^{\frac{4}{3}t} + \frac{4}{3}y(t) e^{\frac{4}{3}t} = -\frac{1}{3}\delta(t) e^{\frac{4}{3}t} = -\frac{1}{3}\delta(t)$$

$$\frac{d}{dt} (y(t) e^{\frac{4}{3}t}) = -\frac{1}{3}\delta(t)$$

Integrating,

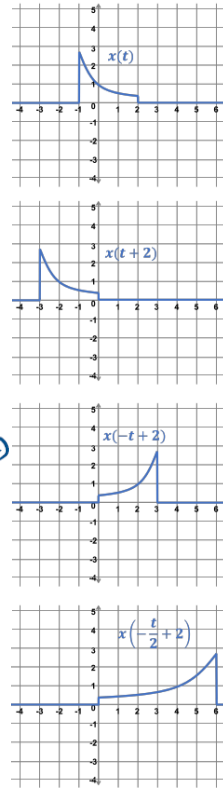
$$y(t) e^{\frac{4}{3}t} \Big|_{0^-}^t = -\int_{0^-}^t \frac{1}{3}\delta(t) dt = -\frac{1}{3} \text{ for } t > 0 \text{ and } 0 \text{ otherwise.}$$

$$y(t) e^{\frac{4}{3}t} - y(0^-) = -\frac{1}{3} u(t)$$

$$y(t) e^{\frac{4}{3}t} = -\frac{1}{3} u(t)$$

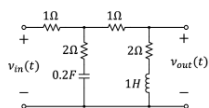
$$y(t) = -\frac{1}{3} e^{-\frac{4}{3}t} \text{ for } t > 0 \text{ and } 0 \text{ otherwise.}$$

Therefore, $y(t) = -\frac{1}{3} e^{-\frac{4}{3}t} u(t)$



a. (25 points) Circuit Analysis

For the circuit below, the input $v_{in}(t) = 1V$ for $t < 0$ and $v_{in}(t) = te^{-2t}u(t)$ for $t > 0$.



a. (5 points) Draw the s-domain representation of the circuit with the initial conditions indicated.

Initial conditions:

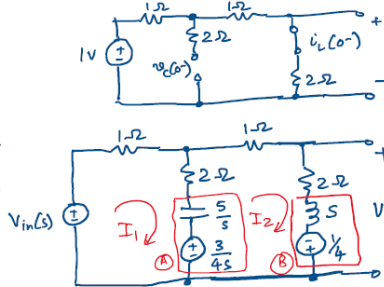
$$i_L(0^-) = 0.25A$$

$$v_C(0^-) = 0.75V$$

s-domain circuit

Grading: .1 pt trying

.+1 pt each (A), (B)



b. (5 points) Find the s-domain representation of the input $v_{in}(t)$ for $t > 0$.

$$v_{in}(t) = te^{-2t}u(t)$$

$$v_{in}(s) = \frac{1}{(s+2)^2}$$

c. (10 points) Write the mesh current equations in the s-domain needed to find the output $V_{out}(s)$. Express your answer by writing out the entries in the Cramer's method matrix shown below in the Laplace domain. Here, I_1 and I_2 represent the mesh currents (left to right), and V_1 , V_2 represent voltages.

$$\begin{pmatrix} A_1 & A_2 \\ A_3 & A_4 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$$

$$\begin{aligned} \text{KVL: } I_1 \left(1 + 2 + \frac{5}{s} \right) - I_2 \left(2 + \frac{5}{s} \right) &= v_{in} - \frac{3}{4s} \\ -I_1 \left(2 + \frac{5}{s} \right) + I_2 \left(\frac{5}{s} + 2 + 1 + s + 2 \right) &= \frac{3}{4s} + \frac{1}{4} \end{aligned}$$

d. (5 points) Write the formula for $V_{out}(s)$ in terms of $V_{in}(s)$. Your answer should be written in terms of A_1 , A_2 , A_3 , A_4 , I_1 , I_2 , $V_{in}(s)$, $V_{out}(s)$.

$$\begin{aligned} \Delta &= A_1 A_4 - A_3 A_2 \Rightarrow I_2 = \frac{\Delta I_2}{\Delta} \\ \Delta I_2 &= A_1 V_2 - A_3 V_1 = \frac{A_1 V_2 - A_3 V_1}{A_1 A_4 - A_3 A_2} \\ V_{out}(s) &= (s+2) I_2 - \frac{1}{4} \end{aligned}$$

4. (15 points) System Response Partitions

A linear time-invariant system is described by the following differential equation:

$$\frac{d^2 y(t)}{dt^2} + 7 \frac{dy(t)}{dt} + 10 y(t) = x(t)$$

The input is $x(t) = e^{-t}u(t)$, with the initial conditions $y(0^-) = 0$, $y'(0^-) = 2$.

a. (5 points) Derive a formula for the output $Y(s)$. Your answer should be expressed as a ratio of polynomials in s with a fully factored denominator.

$$\begin{aligned} y(t) &\xrightarrow{\mathcal{L}} Y(s) \\ \frac{d}{dt} y(t) &\xrightarrow{\mathcal{L}} sY(s) - y(0^-) \\ \frac{d^2}{dt^2} y(t) &\xrightarrow{\mathcal{L}} s^2 Y(s) - sy(0^-) - y'(0^-) \end{aligned}$$

Therefore, taking the Laplace transform of the LCCDE yields:

$$s^2 Y(s) - sy(0^-) - y'(0^-) + 7(sY(s) - y(0^-)) + 10Y(s) = X(s)$$

Using the given initial conditions

$$s^2 Y(s) + 7sY(s) + 10Y(s) - 2 = X(s)$$

$$s^2 Y(s) + 7sY(s) + 10Y(s) = X(s) + 2$$

Also,

$$\begin{aligned} x(t) &\xrightarrow{\mathcal{L}} X(s) \\ e^{-t}u(t) &\xrightarrow{\mathcal{L}} \frac{1}{s+1} \end{aligned}$$

Substituting $X(s) = \frac{1}{s+1}$

$$Y(s)(s^2 + 7s + 10) = \frac{1}{s+1} + 2 = \frac{1+2s+2j}{(s+j)}$$

$$Y(s) = \frac{2s+1+2j}{(s+j)(s^2+7s+10)} = \frac{2s+1+2j}{(s+j)(s+5)(s+2)}$$

$$Y(s) = \frac{2s+1+2j}{(s+j)(s+5)(s+2)}$$

b. (5 points) Find the modes of the system represented by the above LCCDE. Calculate the forced response of this system in the Laplace domain.

The modes of the system correspond to the roots of the characteristic equation. The characteristic equation is: $s^2 + 7s + 10 = 0$ or $(s+5)(s+2) = 0$ or $s = -5, -2$

The partial fraction expansion can be written as

$$Y(s) = \frac{2s+1+2j}{(s+j)(s+5)(s+2)} = \frac{A}{s+j} + \frac{B}{s+5} + \frac{C}{s+2}$$

The natural response corresponds to the modes of the system.

The forced response corresponds to the other term, namely

$$Y_{forced}(s) = \frac{A}{s+j}$$

Finding A,

$$A = \frac{1}{(5-j)(2-j)}$$

$$Y_{forced}(s) = \frac{1}{(5-j)(2-j)(s+j)}$$

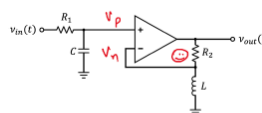
c. (5 points) Calculate the final value $y(\infty)$ of this system's response.

To find the final value, we typically use the final value theorem.

However, one of the poles of the system $\frac{A}{s+j}$ is on the imaginary axis.

This means that the signal does not asymptotically converge to a final fixed value. It is oscillatory as $t \rightarrow \infty$.

5. (10 points) Operational Amplifiers



a. (5 points) Calculate the transfer function $H(s)$ for this circuit.

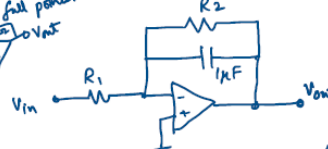
$$v_p = v_n \Rightarrow \frac{1/sC}{R_1 + 1/sC} \times v_{in} = \frac{sL}{R_2 + sL} v_{out}$$

$$\Rightarrow H(s) = \frac{v_{out}(s)}{v_{in}(s)} = \frac{1}{1 + sCR_1} \times \frac{(R_2 + sL)}{sL} = \frac{R_2}{sL(1 + sCR_1)} + \frac{1}{1 + sCR_1}$$

Grading: .1 pt attempt
+1 pt correct
-1 pt H(s)

b. (5 points) Sketch an op-amp circuit that implements the transfer function $H(s) = \frac{-10}{s+2}$. Any capacitor used in your circuit should have a capacitance $C = 1\mu F$.

Notes:
Correct alternate solutions received full points.
e.g. $v_{in} = 2e^{-t}u(t)$



$$\begin{aligned} A = -10 &= \frac{-1}{R_1 \times 10^{-6}} \Rightarrow R_1 = 10^5 \Omega = 100k\Omega \\ P = -2 &= \frac{-1}{R_2 \times 10^{-6}} \Rightarrow R_2 = 500k\Omega \end{aligned}$$

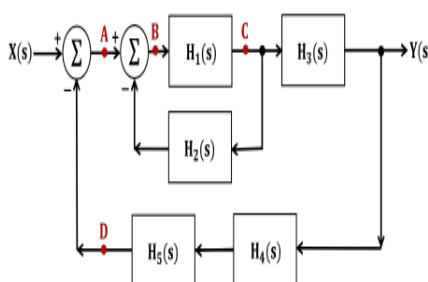
single-pole circuit

Grading:
.1 pt attempt
+1 pt correct circuit
5 pts fully correct answer

6. (10 points) System Block Diagrams

Calculate the overall transfer function $H(s)$ for the following connection of LTI systems. Your final answer

should be expressed in the Laplace domain.



$$C = AH_1 - CH_1H_2$$

$$C = A \frac{H_1}{1 + H_1H_2}$$

$$A = X - YH_4H_5$$

$$C = (X - YH_4H_5) \frac{H_1}{1 + H_1H_2}$$

$$Y = (X - YH_4H_5) \frac{H_1}{1 + H_1H_2} H_3$$

$$Y(1 + H_1H_2) + YH_4H_5H_1H_3 = XH_1H_3$$

$$Y = X \frac{H_1H_3}{(1 + H_1H_2) + H_4H_5H_1H_3}$$

$$\frac{Y}{X} = H = \frac{H_1H_3}{(1 + H_1H_2) + H_4H_5H_1H_3}$$

$$A = X - D \quad (A)$$

$$B = A - CH_2 \quad (B)$$

$$C = BH_1 \quad (C)$$

$$Y = CH_3 \quad (D)$$

$$D = YH_4H_5 \quad (E)$$

Natural / forced partition

Partitioning the total response into a

- **Natural response**: all terms that vary with t matching the system's modes (the example has only one mode: e^{-3t}), and
- **Forced response**: all remaining terms, of the response equation,

the system equation becomes

$$v(t) = \underbrace{[2e^{-3t} + 12e^{-3t}] u(t)}_{\text{Natural response}} + \underbrace{[6 - 18e^{-2t}] u(t)}_{\text{Forced response}}$$

Recall that the exponent $a_c = 3 = 1/RC$, which reflects the natural identity of the circuit. Hence, all terms of the total response that vary as e^{-3t} are regarded as elements of the natural response. Remaining terms are part of the forced response, which includes modes that mimic the character of the excitation function.

Transient / steady-state partition

In many practical situations, the system analyst or designer may be particularly interested in partitioning the response into a

- **Transient response** = all terms that decay to zero as $t \rightarrow \infty$, and a
- **Steady-state response** = all terms that remain after the demise of the transient response.

Partitioning the expression for $v(t)$ along those lines gives

$$v(t) = \underbrace{[14e^{-3t} - 18e^{-2t}] u(t)}_{\text{Transient response}} + \underbrace{6u(t)}_{\text{Steady-state response}}$$

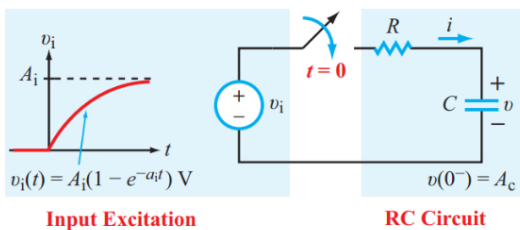
The terms included in the first square bracket decay exponentially to zero as $t \rightarrow \infty$, leaving behind the steady-state component $6u(t)$. The steady-state component need not be a constant (dc value). In fact, if the input function is a sinusoid, the steady-state response will also be a sinusoid at the same frequency.

Zero-state / zero-input partition

As a working example, consider the RC circuit shown below. Prior to closing the switch at $t = 0$, the capacitor voltage was

$$v(0^-) = A_c.$$

Figure 3.14.1: The RC circuit and input excitation are treated as separate entities that get connected at $t = 0$.



Feedback?

Application of KVL at $t \geq 0$ gives

$$RI + v = v_i(t),$$

where v is the voltage across the capacitor and $v_i(t)$ is the input voltage excitation introduced at $t = 0$. Since $i = C dv/dt$ for the capacitor, the equation above can be converted into a first-order differential equation in one variable, ,

$$RC \frac{dv}{dt} + v = v_i(t).$$

To distinguish between the attributes of the RC circuit and those of the input voltage:

- Subscript c refers to attributes that pertain to the circuit alone, independently of the character of the input excitation.
- Subscript i refers to attributes specific to the input excitation.

Accordingly, the initial voltage across the capacitor is

$$v(0^-) = A_c$$

and the characteristic time constant of the RC circuit is

$$\tau = RC = \frac{1}{a_c}.$$

Transforming the system LCCDE $RC \frac{dv}{dt} + v = v_i(t)$ to the s -domain entails

$$\begin{aligned} v(t) &\Rightarrow \mathbf{V}(s) & v_i(t) &\Rightarrow \mathbf{V}_i(s) \\ \frac{dv}{dt} &\Rightarrow s\mathbf{V}(s) - v(0^-) = s\mathbf{V}(s) - A_c \end{aligned}$$

which leads to

$$\underbrace{\left(\frac{s}{a_c} + 1\right) \mathbf{V}(s)}_{\text{Zero state}} = \underbrace{\frac{A_c}{a_c}}_{\text{Initial condition}} + \underbrace{\frac{\mathbf{V}_i(s)}{s + a_c}}_{\text{Input excitation}}.$$

A system (the RC circuit in the present case) is said to be in **zero state** if the initial conditions are zero, i.e., no stored energy. The term on the left-hand side of the equation represents the zero-state status of the circuit. The left side is the s -domain sum of the voltages across the resistor and the capacitor, excluding the initial voltage $v(0^-)$. The first term on the right-hand side (with subscript c) represents the initial conditions as an *equivalent excitation source*, and the last term is the real excitation voltage $\mathbf{V}_i(s)$.

The goal is to solve for $\mathbf{V}(s)$. Dividing all terms in the equation by the factor multiplying $\mathbf{V}(s)$ leads to the expression

$$\underbrace{\mathbf{V}(s)}_{\text{Total response}} = \underbrace{\frac{A_c}{s + a_c}}_{\text{Zero-input response}} + \underbrace{\frac{a_c \mathbf{V}_i(s)}{s + a_c}}_{\text{Zero-state response}}.$$

A system (the RC circuit in the present case) is said to be in **zero state** if the initial conditions are zero, i.e., no stored energy. The term on the left-hand side of the equation represents the zero-state status of the circuit. The left side is the s -domain sum of the voltages across the resistor and the capacitor, excluding the initial voltage $v(0^-)$. The first term on the right-hand side (with subscript c) represents the initial conditions as an *equivalent excitation source*, and the last term is the real excitation voltage $\mathbf{V}_i(s)$.

The goal is to solve for $\mathbf{V}(s)$. Dividing all terms in the equation by the factor multiplying $\mathbf{V}(s)$ leads to the expression

$$\underbrace{\mathbf{V}(s)}_{\text{Total response}} = \underbrace{\frac{A_c}{s + a_c}}_{\text{Zero-input response}} + \underbrace{\frac{a_c \mathbf{V}_i(s)}{s + a_c}}_{\text{Zero-state response}}.$$

The total response is the sum of:

- **Zero-input response (ZIR)** = response of the system in the absence of an input excitation ($\mathbf{V}_i(s) = 0$).
- **Zero-state response (ZSR)** = response of the system for zero state (zero initial conditions, $A_c = v(0^-) = 0$).

Here, the RC circuit is excited by input voltage

$$v_i(t) = A_i[1 - e^{-a_i t}] u(t) \text{ V}$$

with amplitude A_i and exponential coefficient a_i . The corresponding s -domain input expression is

$$\mathbf{V}_i(s) = \frac{A_i}{s} - \frac{A_i}{s + a_i} = \frac{A_i a_i}{s(s + a_i)}.$$

Inserting $\mathbf{V}_i(s)$ into the total-response equation leads to

$$\mathbf{V}(s) = \frac{A_c}{s + a_c} + \frac{A_i a_i}{s(s + a_c)(s + a_i)}.$$

In preparation for transforming the expression to the time domain, express the second term as a sum of partial fractions:

$$\mathbf{V}(s) = \frac{A_c}{s + a_c} + \left[\frac{A_i}{s} + \frac{A_i a_i}{(a_c - a_i) \left(\frac{1}{s + a_c} \right)} + \frac{A_i a_c}{(a_i - a_c) \left(\frac{1}{s + a_i} \right)} \right].$$

Using entries #2 and 3 in the transform pairs table,

$$v(t) = \underbrace{A_c e^{-a_c t} u(t)}_{\text{ZIR}} + \underbrace{\left[A_i + \left(\frac{A_i a_i}{a_c - a_i} \right) e^{-a_c t} + \left(\frac{A_i a_c}{a_i - a_c} \right) e^{-a_i t} \right] u(t)}_{\text{ZSR}}.$$

For $A_c = v(0^-) = 2 \text{ V}$, $a_c = 3$, $A_i = 6 \text{ V}$, and $a_i = 2$,

$$v(t) = \underbrace{2e^{-3t} u(t)}_{\text{ZIR}} + \underbrace{[6 + 12e^{-3t} - 18e^{-2t}] u(t)}_{\text{ZSR}}.$$

The function $e^{-a_c t} = e^{-3t}$ is called a **mode** of the system. The **mode** is independent of the input because $a_c = 1/(RC)$ is characteristic of the RC circuit alone. The zero-input response is composed exclusively of modes, whereas the zero-state response includes the modes of the system as well as other modes that mimic the character of the excitation function. The second term in the equation includes a term that varies as e^{-3t} and two others with functional forms similar to those of the input $v_i(t) = A_i[1 - e^{-a_i t}] u(t) \text{ V}$.